LIBBY — ACCESS GUIDE

Access guide — gbm-mtap-cdkn2a-idhwt-subtotal-c4bq

How to access each therapy: trial contacts + manufacturer medical-info

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How a patient or treating team could practically access each intervention in this case's dossier. Trial recruitment contacts and manufacturer medical-information lines are captured for direct outreach. The number in the first column of the summary table is the entry's identifier in the per-intervention sections below — use it to find the deep section quickly. Information ages — each row carries a Verified date; re-screen before relying on a specific contact or trial slot.

Summary

#	Intervention	Modality	Access status	Regulatory	Recommended first action
1	Bevacizumab (Avastin)	monoclonal_antibody	Standard of care + Off-label use	FDA-approved (full approval, December 2017) for adult recurrent glioblastoma. EMA: not approved for GBM (the EU approval was withdrawn). Multiple biosimilars approved (Mvasi, Zirabev) — covered where bevacizumab is covered.	Carry the intratumoral-hemorrhage flag forward in the chart — when bevacizumab is considered at recurrence, the discussion should include a baseline MRI to confirm no active blood products and a frank risk conversation.
2	Carmustine wafer (Gliadel; polifeprosan 20 with carmustine implant)	device_drug_combination	Standard of care + Off-label use + Clinical trial	FDA-approved for newly diagnosed high-grade glioma as an adjunct to surgery and radiation, and for recurrent GBM as an adjunct to re-resection. EU / EMA: comparable approvals.	For this patient, Gliadel is essentially a retrospective question — the moment of decision (the initial craniotomy) is in the rearview mirror. Not worth pursuing as a stand-alone option.
3	Lomustine (CCNU; Gleostine)	small_molecule	Standard of care + Off-label use	FDA-approved as Gleostine for primary and metastatic brain tumors and for Hodgkin lymphoma. NextSource Biotechnology holds the only FDA-approved version of US-market lomustine.	Resolve MGMT methylation status — same gate as the TMZ decision. If methylated, raise CeTeG with the neuro-oncologist as a Stupp alternative.

4	Temozolomide + radiotherapy (Stupp regimen)	small_molecule	Standard of care + Off-label use	FDA-approved for newly diagnosed glioblastoma multiforme concomitantly with radiotherapy and then as maintenance treatment (Stupp regimen, 2005). Generic since 2014; multiple manufacturers. Brand Temodar marketed by Merck Sharp & Dohme.	Order MGMT promoter methylation testing (MSP or pyrosequencing) on the resection block — this is the single load-bearing question for whether to commit to 6 cycles of adjuvant TMZ.
5	Tumor-Treating Fields (Optune Gio, 200 kHz)	device	Standard of care	FDA-approved (PMA P100034 supplement, October 2015) as Optune in combination with adjuvant temozolomide for newly diagnosed supratentorial GBM after maximal debulking surgery and RT + concurrent TMZ. Rebranded Optune Gio for the GBM indication in 2024 to distinguish from Optune Lua (mesothelioma / NSCLC).	Discuss Optune Gio with the patient during the maintenance-TMZ planning conversation; she needs to understand the adherence commitment before agreeing.
6	Abemaciclib (Verzenio) — CDK4/6 inhibitor	small_molecule	Off-label use + Clinical trial	FDA-approved (2017) for HR+/HER2- advanced or metastatic breast cancer in combination with aromatase inhibitor or fulvestrant, and for adjuvant treatment of HR+/HER2-, node-positive, high-risk early breast cancer. No GBM indication.	Discuss abemaciclib off-label with the neuro-oncologist; if the team is interested, document CDKN2A homozygous loss in the chart and prepare a medical-exception letter citing the Tien 2019 ribociclib phase 0 and the INSIGhT abemaciclib pharmacodynamic data.

7	Dabrafenib (Tafinlar) + Trametinib (Mekinist)	small_molecule_combination	Off-label use	FDA-approved (June 2022) as a tumor-agnostic indication for unresectable or metastatic BRAF V600E-mutant solid tumors in adults and children >=6 with no satisfactory alternative treatment. Multiple individual-tumor approvals (melanoma, NSCLC, anaplastic thyroid, etc.).	Order BRAF V600E testing (IHC with VE1 antibody or NGS) on the resection block — this is a known diagnostic gap from profile.json.
8	Entrectinib (Rozlytrek) — TRK / ROS1 / ALK inhibitor	small_molecule	Off-label use	FDA-approved (August 2019) for adult and pediatric (>=12) patients with NTRK fusion-positive solid tumors that are metastatic or where surgery would cause severe morbidity, with no satisfactory alternative or post-progression. Also approved for ROS1-positive metastatic NSCLC. EMA-approved with similar wording.	Order RNA-based fusion NGS (same single test resolves both larotrectinib and entrectinib eligibility).
9	Larotrectinib (Vitrakvi) — NTRK inhibitor	small_molecule	Off-label use	FDA-approved (November 2018 accelerated, April 2025 full approval) as a tumor-agnostic indication for solid tumors with NTRK gene fusion without known acquired resistance mutation, metastatic or with no satisfactory alternatives or post-progression. Approved by EMA / UK / Japan with similar wording.	Order RNA-based fusion NGS to look for NTRK1/2/3 fusions — this is a known diagnostic gap from profile.json.

10	Nivolumab (Opdivo)	monoclonal_antibody	Off-label use	FDA-approved for multiple solid tumors (melanoma, NSCLC, RCC, classical Hodgkin lymphoma, urothelial, etc.) and for MSI-H/dMMR colorectal cancer (tumor-agnostic in CRC, not broader). NOT approved for GBM. CheckMate-143, -498, -548 all failed to show OS benefit in GBM.	Do not pursue off-label nivolumab in GBM — the failed trials are too well-known to overcome at payer review.
11	Palbociclib (Ibrance) — CDK4/6 inhibitor	small_molecule	Off-label use	FDA-approved (2015) for HR+/HER2- advanced/metastatic breast cancer in combination with endocrine therapy. No GBM indication.	If a CDK4/6 inhibitor is going to be used off-label for this patient, prefer abemaciclib over palbociclib because of CNS penetration and modestly better preclinical / phase 0 GBM data.
12	Pembrolizumab (Keytruda)	monoclonal_antibody	Off-label use + Clinical trial	FDA-approved with multiple tumor-agnostic indications relevant here: (1) MSI-H or dMMR solid tumors post-treatment (full approval Mar 2023); (2) TMB-H (>=10 mut/Mb) solid tumors post-treatment (accelerated approval 2020). EMA-approved for MSI-H/dMMR in five tumor types.	Order MMR IHC (MLH1/MSH2/MSH6/PMS2) and TMB by NGS on the resection block — these are listed diagnostic gaps in profile.json.

13	Ribociclib (Kisqali) — CDK4/6 inhibitor	small_molecule	Off-label use	FDA-approved (2017) for HR+/HER2- advanced/metastatic breast cancer and (2024 expanded) for adjuvant HR+/HER2- early breast cancer in combination with endocrine therapy. No GBM indication.	If a CDK4/6 inhibitor is pursued off-label, prefer abemaciclib first (better GBM-specific clinical data). Ribociclib is the second choice.
14	B7-H3 CAR T (class — pediatric DFCI + adult China protocols)	cell_therapy	Clinical trial + Not yet accessible	Investigational. Academic INDs across multiple sponsors.	File B7-H3 CAR T as a forward-looking option not currently accessible.
15	BGB-58067 (brain-penetrant MTA-cooperative PRMT5 inhibitor)	small_molecule	Clinical trial	IND-cleared; no approval. Investigational only.	Email or call the Phase 0 Navigator at 602-406-8605 to ask about MTAP-IHC slot availability and current screening capacity in both Arm A and Arm B.
16	CAN-3110 (rQNestin34.5v.2) — oncolytic HSV-1	oncolytic_virus	Clinical trial	Investigational; FDA Orphan Drug Designation for recurrent high-grade glioma; FDA Fast Track Designation for recurrent HGG. Candel Therapeutics.	File CAN-3110 in the post-progression plan, not the front-line plan. Patient must have documented progression on Stupp before she becomes eligible.
17	CARv3-TEAM-E (MGH — CAR T secreting EGFR x CD3 bispecific T-cell engager)	cell_therapy	Clinical trial	Investigational. Academic IND, Massachusetts General Hospital.	Order MGMT methylation + EGFRvIII testing on the resection block in parallel — these are the two load-bearing biomarkers for INCIPIENT Arm 2 eligibility.
18	DCVax-L — autologous dendritic-cell vaccine pulsed with tumor lysate	cellular_vaccine	Clinical trial + Expanded access program	Investigational in the US (no BLA filed). UK MHRA Marketing Authorization Application submitted December 2023 under 150-day fast-track review for unmet medical needs; decision expected late 2025 / early 2026. EMA not yet filed.	Email marnix@nwbio.com to confirm there is no new-patient enrollment route through the existing US protocols. Do NOT raise the patient's hopes about DCVax-L until that confirmation is in hand.

19	DNX-2401 (tasadenoturev / Delta-24-RGD) — oncolytic adenovirus	oncolytic_virus	Clinical trial	Investigational. DNAtrix (Houston, TX). FDA Orphan Drug Designation for HGG.	Treat DNX-2401 as forward-looking — call DNAtrix general inquiry (via the dnatrix.com website) to ask whether a successor trial is in planning.
20	Dual-targeting CAR-NK cells (Beijing Biotech)	cell_therapy	Clinical trial + Not yet accessible	Investigational. China-only IND.	Do not pursue — geography mismatch.
21	E-SYNC — UCSF anti-EGFRvIII synNotch -> anti-EphA2/IL-13Ra2 CAR T cells	cell_therapy	Clinical trial	Investigational. Academic IND, UCSF.	Order EGFRvIII testing — diagnostic gap from profile.json.
22	EGFRvIII-directed CAR T cells (class — multiple constructs)	cell_therapy	Clinical trial	Investigational. Multiple academic INDs across Penn, NCI, MGH, Duke, UCSF. No FDA approval.	Order EGFRvIII testing (RT-PCR or NGS fusion call) on the resection block — this is a known diagnostic gap from profile.json.
23	Focused ultrasound BBB opening (SonoCloud-9 / NaviFUS / Sonabend FUS programs)	device	Clinical trial + Not yet accessible	Investigational. FDA Breakthrough Device Designation for SonoCloud-9 (2022). FDA / EMA Orphan Drug Designation 2023 for carboplatin + SonoCloud. No US device approval.	File FUS-BBB programs in the post-progression plan.
24	GBM AGILE — adaptive platform trial (multi-arm)	platform_trial	Clinical trial	Multiple investigational arms; standard-of-care control arm. Platform run by Global Coalition for Adaptive Research (GCAR).	Call GCAR patient info at 310-598-3199 or email patientinfo@gcaresearch.org to identify the nearest GBM AGILE site to the patient.
25	IDE397 — MAT2A inhibitor (IDEAYA)	small_molecule	Clinical trial + Not yet accessible	IND-cleared; no approval. IDEAYA Biosciences.	Do not pursue NCT04794699 enrollment — primary CNS malignancy is an exclusion criterion.
26	IDE892 — next-generation MTA-cooperative PRMT5 inhibitor (IDEAYA)	small_molecule	Clinical trial + Not yet accessible	IND-cleared 2025; no approval. IDEAYA Biosciences.	Do not pursue NCT07277413 — primary CNS exclusion.

27	Lerapolturev (PVSRIPO) — recombinant oncolytic poliovirus	oncolytic_virus	Clinical trial	Investigational. FDA Breakthrough Therapy Designation for recurrent GBM. FDA Orphan Drug Designation for melanoma and GBM. Istari Oncology.	File PVSRIPO in the post-progression plan. Document polio vaccination status now; if the patient hasn't had a recent booster, schedule one — it's a protocol requirement for LUMINOS-101 enrollment.
28	M032 — oncolytic HSV-1 expressing IL-12	oncolytic_virus	Clinical trial	Investigational. University of Alabama Birmingham program (Friedman / Markert).	Call UAB neurosurgery (the published M032 program contacts are Burt Nabors and Renee Chambers) to ask whether any M032 adult GBM trial is currently enrolling.
29	MRTX1719 / Navlimetostat (BMS-986504) — MTA-cooperative PRMT5 inhibitor	small_molecule	Clinical trial	IND-cleared; no approval. Bristol Myers Squibb development program (acquired with Mirati in 2023).	Call BMS Clinical Trials Contact Center at 855-907-3286 to confirm whether NCT05245500 will accept a primary brain tumor (GBM) under the existing protocol amendments.
30	OH2 — oncolytic HSV-2 (Binhui)	oncolytic_virus	Clinical trial + Not yet accessible	Investigational. China-only.	Do not pursue — geography mismatch.
31	Penn (Bagley bivalent intrathecal CAR T)	cell_therapy	Clinical trial	Investigational. Academic IND, University of Pennsylvania.	Order EGFR amplification testing on the resection block (FISH or NGS copy-number) — diagnostic gap from profile.json.
32	S095035 — MAT2A inhibitor (Servier)	small_molecule	Clinical trial + Not yet accessible	IND-cleared; no approval. Servier Bio-Innovation.	Call Servier medical information at 1-800-770-0631 and ask specifically whether NCT06188702 will re-open enrollment for the IDHwt GBM cohort.
33	SurVaxM — survivin peptide vaccine	peptide_vaccine	Clinical trial	Investigational; no approval. MimiVax (developed at Roswell Park).	Call Roswell Park (1-800-ROSWELL / 1-800-767-9355) to ask whether the SURVIVE trial has any open screening capacity at any of the 11 US sites — the registry says active not recruiting but site-level reactivations sometimes happen.
34	TNG462 / Vopimetostat — peripheral MTA-cooperative PRMT5 inhibitor	small_molecule	Clinical trial	IND-cleared; no approval. Tango Therapeutics.	Do not pursue TNG462 enrollment — wrong tumor biology (peripheral compartment).

35	AG-270 / S095033 — first-generation MAT2A inhibitor	small_molecule	Unavailable	Development discontinued. AG-270 was the Agios MAT2A inhibitor (later Servier S095033); the phase 1 study (Sumi 2024 Nat Commun) read out and the program did not advance to phase 2.	Treat AG-270 as foundational evidence only.
36	Anvumetostat (AMG 193) — MTA-cooperative PRMT5 inhibitor	small_molecule	Unavailable	Development discontinued. Amgen announced on its Q1 2026 earnings call (April 30, 2026) that it stopped four phase 1b/2 trials of anvumetostat because efficacy did not meet the company's bar for additional internal investment.	Do NOT pursue anvumetostat enrollment — the program is closed.
37	Depatuxizumab mafodotin (Depatux-M / ABT-414) — EGFR-directed ADC	antibody_drug_conjugate	Unavailable	Development discontinued. AbbVie halted Phase 3 INTELLANCE-1 in 2019 after the IDMC concluded the addition of depatux-m to RT + TMZ did not improve OS over placebo in EGFR-amplified newly diagnosed GBM. All ongoing depatux-m studies were halted at the same time.	Treat depatux-m as a closed door.
38	Rindopepimut (CDX-110) — EGFRvIII peptide vaccine	peptide_vaccine	Unavailable	Development discontinued. Phase 3 ACT IV (newly diagnosed EGFRvIII+ GBM) terminated for futility in 2016; median OS 20.4 mo (rindopepimut) vs 21.1 mo (control). All ongoing studies were closed.	Treat rindopepimut as a closed door.

39	TNG908 — brain-penetrant MTA-cooperative PRMT5 inhibitor	small_molecule	Unavailable	Investigational, no approval. Trial terminated.	Treat TNG908 as a closed door.
40	Toca 511 + Toca FC (vocimagene amiretrorepvec + 5-fluorocytosine)	gene_therapy	Unavailable	Development discontinued. Phase 3 Toca 5 trial (NCT02414165) failed primary OS endpoint in 2019 (11.1 vs 12.2 mo). Tocagen restructured (~65% staff reduction) and effectively closed the program. The continuation protocol NCT04327011 supported only previously-enrolled patients.	Treat Toca 511 as a closed door.

Standard of care (5)

1. Bevacizumab (Avastin)

Aliases: bevacizumab, Avastin, bevacizumab-awwb, bevacizumab-bvzr, Mvasi, Zirabev

Modality: monoclonal_antibody · **Regulatory:** FDA-approved (full approval, December 2017) for adult recurrent glioblastoma. EMA: not approved for GBM (the EU approval was withdrawn). Multiple biosimilars approved (Mvasi, Zirabev) — covered where bevacizumab is covered. · **Guidelines:** NCCN category 2A for recurrent GBM as monotherapy or with lomustine. Not standard for newly diagnosed (AVAglio and RTOG 0825 both negative for OS). · **Geographic scope:** US: bevacizumab on-label for recurrent GBM. EU: GBM approval was withdrawn — for EU patients this becomes off-label. · **Verified:** 2026-05-26

Bevacizumab is on-label and accessible for the recurrent setting whenever the patient progresses. The load-bearing constraint for THIS patient is the intratumoral hemorrhage at presentation — bevacizumab is associated with intracranial hemorrhage risk, and the package insert carries a serious-bleeding warning. The hemorrhage history is not an absolute contraindication, but the treating team should weigh it carefully and ideally document a stable post-resection MRI before any anti-VEGF exposure.

Next steps

1. Carry the intratumoral-hemorrhage flag forward in the chart — when bevacizumab is considered at recurrence, the discussion should include a baseline MRI to confirm no active blood products and a frank risk conversation.
2. For the front-line phase, bevacizumab is NOT indicated (AVAglio / RTOG 0825 negative); do not let payers approve it for newly diagnosed disease just because the patient is symptomatic.
3. If steroid-sparing for edema management becomes urgent, bevacizumab can be considered earlier but the

hemorrhage history shifts the risk/benefit.
4. Biosimilars (Mvasi, Zirabev) are interchangeable for the GBM indication — let the payer pick the cheapest one.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT00345163	2	completed	no	—

Manufacturer / sponsor contact

- **Company:** Genentech (Roche)
- **Med info phone:**1-800-821-8590
- **Product info:**https://www.gene.com/medical-professionals/medicines/avastin
- **Compassionate / expanded access:**https://www.gene.com/patients/genentech-access-solutions
- **Notes:** Genentech Medicine Information Support: (800) 821-8590 (M-F 5am-5pm PT). Avastin Access Solutions (case management, benefits investigation): 1-888-249-4918. Adverse-event reporting: 1-888-835-2555. Biosimilars (Amgen's Mvasi, Pfizer's Zirabev) are typically cheaper and often preferred by payers.

Payer / coverage: On-label for recurrent GBM — Medicare and major commercial plans cover with prior auth. The patient is currently newly diagnosed, so any front-line bevacizumab use (e.g. AVAglio-style addition to Stupp) is off-label and faces payer pushback. Biosimilar substitution is now common and reduces cost ~25-50%.

Notes: AVAglio (Chinot 2014 NEJM) and RTOG 0825 (Gilbert 2014 NEJM) both showed PFS but not OS benefit when bevacizumab was added to Stupp for newly diagnosed GBM. EORTC 26101 (Wick 2017 NEJM) showed lomustine + bevacizumab improves PFS but not OS over lomustine alone in recurrent disease. The 2017 full FDA approval is based on the Friedman / Kreisl single-arm data.

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2. Carmustine wafer (Gliadel; polifeprosan 20 with carmustine implant)

Aliases: Gliadel, carmustine wafer, BCNU wafer, polifeprosan 20

Modality: device_drug_combination · **Regulatory:** FDA-approved for newly diagnosed high-grade glioma as an adjunct to surgery and radiation, and for recurrent GBM as an adjunct to re-resection. EU / EMA: comparable approvals. · **Guidelines:** NCCN category 2B for newly diagnosed high-grade glioma and for recurrent GBM (lower than category 1 because the modern data are mixed and many neuro-oncology programs no longer use it routinely). · **Geographic scope:** US, EU, UK, AU all carry the approval. · **Verified:** 2026-05-26

Gliadel is technically available but its use has declined sharply over the last decade — most large neuro-oncology centers no longer place wafers routinely because the absolute OS benefit is small (~2 mo) and wafer-related complications (cerebral edema, healing problems, infection) can compromise the patient's ability to start adjuvant RT on schedule. For THIS patient, the moment for wafer placement has already passed (her subtotal resection happened without wafers); a reopen-and-place isn't realistic unless she has a second-look resection.

Next steps

1. For this patient, Gliadel is essentially a retrospective question — the moment of decision (the initial craniotomy) is in the rearview mirror. Not worth pursuing as a stand-alone option.

2. If the patient has a planned re-resection at recurrence, the question of Gliadel at THAT operation is reasonable to raise with the surgeon — but most centers prefer to enroll on a trial (CAN-3110, PVSRIPO, MGH CARv3-TEAM-E intracavitary protocols) rather than place wafers.
3. If the team is considering the Johns Hopkins NCT05083754 trial, the patient would need a second resection that includes wafer placement — Kleinberg's group (410-614-2597) screens for the protocol.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT05083754	1/2	recruiting	no	Lawrence Kleinberg, MD; kleinla@jhmi.edu; 410-614-2597

Manufacturer / sponsor contact

- **Company:** Azurity Pharmaceuticals (acquired Arbor Pharmaceuticals; manufactured by Eisai for Arbor/Azurity at Baltimore facility)
- **Med info phone:**1-866-516-4950
- **Product info:**https://gliadel.com/
- **Notes:** Arbor Pharmaceuticals Medical Information: 1-866-516-4950. Adverse-event reporting Azurity: 1-800-461-7449. Gliadel must be placed during craniotomy — the surgeon orders the wafers in advance based on planned resection cavity size.

Payer / coverage: Medicare and commercial plans cover the wafers when placed during an FDA-on-label resection. Each procedure typically uses 6-8 wafers (~$15-20K each), so total drug cost is meaningful. Prior auth is procedure-bundled with the craniotomy.

Notes: Gliadel was approved in 1996 (newly diagnosed) and 2003 (recurrent). The drug has changed hands multiple times: Guilford -> MGI Pharma -> Eisai -> Arbor Pharmaceuticals (2012) -> Azurity (Azurity acquired Arbor). Manufacturing remains at the Eisai Baltimore facility under long-term supply agreement.

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3. Lomustine (CCNU; Gleostine)

Aliases: lomustine, CCNU, Gleostine, CeeNU

Modality: small_molecule · **Regulatory:** FDA-approved as Gleostine for primary and metastatic brain tumors and for Hodgkin lymphoma. NextSource Biotechnology holds the only FDA-approved version of US-market lomustine. · **Guidelines:** NCCN category 1 for recurrent GBM, both as monotherapy and as the CCNU + bevacizumab combination from EORTC 26101. The CeTeG / NOA-09 regimen (CCNU + TMZ for newly diagnosed MGMT-methylated GBM) is NCCN-listed as a category 2A alternative to Stupp in MGMT-methylated tumors. · **Geographic scope:** Globally available; NextSource is the sole US-market manufacturer. · **Verified:** 2026-05-26

Lomustine is on-label and accessible — but the load-bearing decision is whether to use it. If MGMT comes back methylated, the CeTeG / NOA-09 regimen (CCNU + TMZ + RT) is a reasonable Stupp alternative with better median OS in that subgroup, at the cost of heavier cytopenia. For recurrent disease later, CCNU is the standard backbone (alone or with bevacizumab if the hemorrhage history is no longer disqualifying).

Next steps

1. Resolve MGMT methylation status — same gate as the TMZ decision. If methylated, raise CeTeG with the neuro-oncologist as a Stupp alternative.
2. If the patient ultimately gets CCNU, monitor CBC q2 weeks for the first cycle; the nadir is delayed (~4-6 weeks) and easy to miss.
3. For uninsured / underinsured patients call NextSource Cares at 877-438-9759 before the first prescription is written.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT01149109	3	completed	unconfirmed	—
NCT01290939	3	completed	no	—

Manufacturer / sponsor contact

- **Company:** NextSource Biotechnology
- **Med info phone:**855-672-2468
- **Med info email:**info@nextsourcebio.com
- **Product info:**https://gleostine.com/
- **Compassionate / expanded access:**https://gleostine.com/nextsource-cares/
- **Notes:** Patient-assistance line for free Gleostine when financial need is documented: 877-438-9759 (NextSource Cares). Pricing has been historically controversial — NextSource raised the price ~14x after acquiring rights, which is the load-bearing payer issue for this drug.

Payer / coverage: Covered by Medicare Part D and most commercial plans, but the per-cycle list price is substantial (~$10-20K/cycle depending on dose). For uninsured patients NextSource Cares provides free product when eligibility documentation is approved. Prior auth is rarely an obstacle for the GBM indication.

Notes: Lomustine has been around since the 1970s. The pricing situation is the modern wrinkle — the drug used to be cheap and is now expensive enough that patient-assistance enrollment is sometimes the difference between adherence and skipped doses.

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4. Temozolomide + radiotherapy (Stupp regimen)

Aliases: temozolomide, TMZ, Temodar, Temodal, SCH 52365

Modality: small_molecule · **Regulatory:** FDA-approved for newly diagnosed glioblastoma multiforme concomitantly with radiotherapy and then as maintenance treatment (Stupp regimen, 2005). Generic since 2014; multiple manufacturers. Brand Temodar marketed by Merck Sharp & Dohme. · **Guidelines:** NCCN category 1 (preferred) for newly diagnosed GBM as concurrent RT + TMZ (75 mg/m^2/day x6 weeks) followed by 6 cycles adjuvant TMZ (150-200 mg/m^2 d1-5 q28d). MGMT-methylated tumors derive the largest benefit (Hegi 2005, Stupp 2009 5-year update). · **Geographic scope:** Globally available — US, EU, UK, AU, Japan all carry the GBM indication. · **Verified:** 2026-05-26

TMZ is the chemotherapy backbone of standard-of-care for newly diagnosed GBM and is the easiest piece of the regimen to access. The interesting question is not whether the patient can get TMZ but whether she should — that hinges on MGMT promoter methylation, which is currently unknown in this case. Order the MSP/pyrosequencing assay before locking in the 6-month adjuvant commitment.

Next steps

1. Order MGMT promoter methylation testing (MSP or pyrosequencing) on the resection block — this is the single load-bearing question for whether to commit to 6 cycles of adjuvant TMZ.
2. If methylated: proceed with concurrent RT + TMZ 75 mg/m^2 daily x6 weeks, then adjuvant TMZ 150-200 mg/m^2 d1-5 q28d x6 cycles per Stupp.
3. If unmethylated: discuss with the neuro-oncologist whether to (a) still follow Stupp, (b) try the GBM AGILE control or experimental arms, or (c) consider trials that drop TMZ in favor of an MTAP-directed or immunologic backbone.
4. For elderly-leaning protocols (the patient is 66), consider the Perry hypofractionated 40 Gy/15 fx + TMZ schedule, which has comparable outcomes with less RT burden.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT00006353	3	completed	yes	—
NCT00482677	3	completed	yes	—

Manufacturer / sponsor contact

- **Company:** Merck Sharp & Dohme (Temodar brand); multiple generic manufacturers
- **Med info phone:**1-800-444-2080
- **Product info:**https://www.merck.com/product/usa/pi_circulars/t/temodar/temodar_capsules_pi.pdf
- **Notes:** The Merck National Service Center handles medical-information and adverse-event inquiries for branded Temodar. Generic temozolomide is widely available through standard oncology pharmacies — most patients now receive a generic, and Medicare Part D covers oral chemo.

Payer / coverage: Universally covered for newly diagnosed GBM in the US. Methylation status does not gate coverage — it gates expected benefit. If MGMT comes back unmethylated, the patient and treating team need to weigh whether the marginal absolute benefit (~2 mo OS) is worth the cytopenia risk, or whether they'd rather slot into a TMZ-sparing front-line trial (CheckMate-498 logic, GBM AGILE control alternatives).

Notes: The Stupp regimen has been standard of care for 20 years. CCTG CE.6 / Perry validated the short-course schedule for elderly patients. Stupp 2009 5-year update remains the long-term reference.

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5. Tumor-Treating Fields (Optune Gio, 200 kHz)

Aliases: TTFields, Optune, Optune Gio, NovoTTF-100A, Tumor Treating Fields

Modality: device · **Regulatory:** FDA-approved (PMA P100034 supplement, October 2015) as Optune in combination with adjuvant temozolomide for newly diagnosed supratentorial GBM after maximal debulking surgery and RT + concurrent TMZ. Rebranded Optune Gio for the GBM indication in 2024 to distinguish from Optune Lua (mesothelioma / NSCLC). · **Guidelines:** NCCN category 1 for newly diagnosed GBM in combination with maintenance temozolomide (EF-14 trial; Stupp 2017 JAMA). Recommended after completion of concurrent RT + TMZ. · **Geographic scope:** US, EU, Japan all carry the GBM indication; >600 US centers prescribe Optune. · **Verified:** 2026-05-26

Optune Gio is on-label and routinely accessible for this patient once she finishes the concurrent RT + TMZ phase and moves to maintenance TMZ. The practical hurdles are cosmetic (continuous scalp-array wear), adherence (>=18 hours/day is the threshold for the EF-14 OS benefit), and getting through prior auth — Novocure's support team does the latter.

Next steps

1. Discuss Optune Gio with the patient during the maintenance-TMZ planning conversation; she needs to understand the adherence commitment before agreeing.
2. Have the prescribing center submit the Optune Rx; MyNovocure (1-855-281-9301) takes it from there for benefits investigation and shipping.
3. If the patient declines the wear-time commitment, document the discussion — TTFields is the only newly diagnosed GBM modality besides Stupp that has prospective Phase 3 OS evidence, so the decision matters.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT00916409	3	completed	yes	—
NCT04471844	3	completed	yes	—

Manufacturer / sponsor contact

- **Company:** Novocure
- **Med info phone:**1-855-281-9301
- **Med info email:**support@mynovocure.com
- **Product info:**https://www.optunegio.com/
- **Notes:** MyNovocure support program runs benefits investigation, device shipping, and the device-support-specialist visits. Optune is prescribed through a certified prescriber and the device is shipped to the patient's home. The patient must commit to >=18 hours/day wear time for the OS benefit seen in EF-14.

Payer / coverage: Medicare covers Optune under DME benefit (NCD 110.24 for TTFields in GBM) for newly diagnosed and recurrent GBM; commercial coverage varies but most major plans cover with prior auth requiring documentation of Stupp completion and ECOG/KPS. Novocure's nCompass program handles the prior-auth paperwork.

Notes: Optune was rebranded Optune Gio in 2024 to separate the GBM indication from the mesothelioma/NSCLC Optune Lua product. The underlying device is unchanged.

Off-label use (8)

6. Abemaciclib (Verzenio) — CDK4/6 inhibitor

Aliases: abemaciclib, Verzenio, LY2835219

Modality: small_molecule · **Regulatory:** FDA-approved (2017) for HR+/HER2- advanced or metastatic breast cancer in combination with aromatase inhibitor or fulvestrant, and for adjuvant treatment of HR+/HER2-, node-positive, high-risk early breast cancer. No GBM indication. · **Guidelines:** NCCN category 1 for breast-cancer indications. Off-guideline for GBM. NCCN Compendium does NOT currently list GBM as a compendia-supported off-label use. · **Geographic scope:** Globally available on-label for breast cancer; off-label GBM use depends on local payer. · **Verified:** 2026-05-26

Abemaciclib is the most CNS-penetrant CDK4/6 inhibitor and the patient's CDKN2A homozygous loss gives mechanistic justification, but the GBM evidence remains inconsistent — the INSIGhT abemaciclib + TMZ arm closed without graduating, and the DFCI recurrent-GBM phase 0/2 is closed. Off-label prescription is possible but is a payer fight (no NCCN Compendium support) and the clinical-evidence base for solo abemaciclib in newly diagnosed GBM is thin. Most centers reserve it for trial settings or for compassionate post-progression scenarios. Watch the elderly cytopenia and diarrhea profile in a 66-year-old.

Next steps

1. Discuss abemaciclib off-label with the neuro-oncologist; if the team is interested, document CDKN2A homozygous loss in the chart and prepare a medical-exception letter citing the Tien 2019 ribociclib phase 0 and the INSIGhT abemaciclib pharmacodynamic data.
2. If a clinical trial is preferred, prioritize the BGB-58067 (Phoenix) or MountainTAP-5 (BMS, opening July 2026) PRMT5 axes ahead of abemaciclib — the PRMT5 evidence base is stronger for the patient's MTAP/CDKN2A co-loss biology.
3. For dosing if off-label is pursued, start at 150 mg BID with close diarrhea/cytopenia monitoring; the patient's age (66) raises G3+ diarrhea risk and may warrant dose reduction.
4. Call Lilly Answers Center (1-800-545-5979) for the latest published off-label medical-information dossier; they will provide the published Verzenio CNS data on request.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT02981940	0/2	active not recruiting	no	Eudocia Q Lee, MD MPH; eudocia_lee@dfci.harvard.edu; 617-632-2166
NCT02977780	2	recruiting	no	Patrick Y. Wen, MD; patrick_wen@dfci.harvard.edu; 617-632-2166

Manufacturer / sponsor contact

- **Company:** Eli Lilly and Company
- **Med info phone:**1-800-545-5979
- **Product info:**https://verzenio.lilly.com/hcp
- **Compassionate / expanded access:** https://medical.lilly.com/us/products/answers/is-there-an-expanded-access-or-compassionate-use-program-available-for-ver
- **Notes:** Lilly Answers Center: 1-800-LillyRx. Verzenio support line: 1-844-VERZENIO (1-844-837-9364). Lilly's published position on Verzenio expanded access is that compassionate-use requests are evaluated on a

case-by-case basis when no clinical trial is available — they do NOT operate a published GBM-specific compassionate-use protocol.

Payer / coverage: Off-label GBM prescribing is a real payer fight. The NCCN Compendium does not list GBM, so medical-exception requests rely on the published CDK4/6 + CDKN2A loss preclinical rationale (Tien 2019 phase 0 ribociclib, INSIGhT abemaciclib data) and the patient's confirmed CDKN2A homozygous loss. Expect at least one round of denial. Lilly's PA program (LillyCares) can help uninsured patients access Verzenio for on-label indications, but does not cover off-label GBM use.

Notes: Abemaciclib is the most studied CDK4/6 inhibitor in GBM because of its CNS penetration (Raub 2015). The class story is mixed — palbociclib failed (Taal 2018), ribociclib only had a phase 0 PD/PK readout (Tien 2019). The CDKN2A-loss subgroup is the right mechanistic frame but the prospective data in that subgroup are still pending.

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7. Dabrafenib (Tafinlar) + Trametinib (Mekinist)

Aliases: dabrafenib, trametinib, Tafinlar, Mekinist, GSK2118436, GSK1120212

Modality: small_molecule_combination · **Regulatory:** FDA-approved (June 2022) as a tumor-agnostic indication for unresectable or metastatic BRAF V600E-mutant solid tumors in adults and children >=6 with no satisfactory alternative treatment. Multiple individual-tumor approvals (melanoma, NSCLC, anaplastic thyroid, etc.). · **Guidelines:** NCCN category 1 for BRAF V600E-mutant melanoma and NSCLC. The tumor-agnostic indication is NCCN-recognized when the V600E mutation is documented. Specifically for high-grade glioma, the ROAR trial (Wen 2022 Lancet Oncol) is the supporting evidence. · **Geographic scope:** US: tumor-agnostic FDA approval since 2022. EU: similar approval via EMA centralized procedure. · **Verified:** 2026-05-26

Dabrafenib + trametinib is the cleanest tumor-agnostic access path IF the patient turns out to have a BRAF V600E mutation — but BRAF V600E in adult IDH-wildtype GBM is rare (~1-2%) and the patient's NGS has not reported BRAF status yet. The whole story turns on the biomarker. Order BRAF V600E testing (IHC VE1 or NGS) on the resection block before pursuing this path.

Next steps

1. Order BRAF V600E testing (IHC with VE1 antibody or NGS) on the resection block — this is a known diagnostic gap from profile.json.
2. If BRAF V600E is positive, the prescription path is straightforward; Novartis support handles benefits investigation via 1-800-282-7630.
3. If BRAF V600E is negative, this entire row drops off the dossier. The published GBM data (Wen 2022 ROAR, ORR 33%) is encouraging but the eligibility gate is the rate-limiting question.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT02034110	2	active not recruiting	unconfirmed	—

Manufacturer / sponsor contact

- **Company:** Novartis
- **Med info phone:**1-888-669-6682
- **Product info:**https://www.us.tafinlarmekinist.com/
- **Notes:** Novartis Patient Assistance Now Oncology: 1-800-282-7630. Tafinlar/Mekinist co-pay program: $0/month for eligible commercially-insured patients.

Payer / coverage: Once a BRAF V600E mutation is documented, on-label tumor-agnostic prescribing is well-supported by Medicare and major commercial plans. Prior auth typically requires the molecular report and a no-satisfactory-alternative attestation.

Notes: ROAR's HGG cohort (Wen 2022 Lancet Oncol) is the foundational adult GBM evidence. Pediatric LGG data are larger (Bouffet 2023 NEJM) but the patient is adult.

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8. Entrectinib (Rozlytrek) — TRK / ROS1 / ALK inhibitor

Aliases: entrectinib, Rozlytrek, RXDX-101

Modality: small_molecule · **Regulatory:** FDA-approved (August 2019) for adult and pediatric (>=12) patients with NTRK fusion-positive solid tumors that are metastatic or where surgery would cause severe morbidity, with no satisfactory alternative or post-progression. Also approved for ROS1-positive metastatic NSCLC. EMA-approved with similar wording. · **Guidelines:** NCCN tumor-agnostic indication recognized when NTRK fusion is documented. Comparable to larotrectinib. · **Geographic scope:** US, EU all carry tumor-agnostic NTRK approval. · **Verified:** 2026-05-26

Entrectinib is the second tumor-agnostic NTRK inhibitor and the one with better-documented CNS exposure pharmacology. Same biomarker gate as larotrectinib — the patient needs NTRK fusion testing first. If both are options, most neuro-oncologists prefer entrectinib for CNS disease because of its design for CNS penetration.

Next steps

1. Order RNA-based fusion NGS (same single test resolves both larotrectinib and entrectinib eligibility).
2. If NTRK fusion is positive, entrectinib is typically the CNS-preferred TRK inhibitor; Genentech Access Solutions (866-422-2377) handles benefits investigation.
3. If NTRK testing identifies a ROS1 fusion (rare in glioma but possible), entrectinib is on-label for that as well — though ROS1+ glioma is exceedingly rare in adults.

Manufacturer / sponsor contact

- **Company:** Genentech (Roche)
- **Med info phone:**1-800-821-8590
- **Product info:**https://www.gene.com/medical-professionals/medicines/rozlytrek
- **Compassionate / expanded access:**https://www.gene.com/patients/genentech-access-solutions
- **Notes:** Genentech Access Solutions: (866) 422-2377 for benefits investigation and patient assistance. Adverse-event reporting: 1-888-835-2555. Entrectinib has broader CNS pharmacology data than larotrectinib — the molecule was originally designed for CNS exposure.

Payer / coverage: Same as larotrectinib — on-label when NTRK fusion is documented, well-supported by NCCN Compendium.

Notes: Entrectinib's CNS-penetrant design is the key clinical differentiator vs larotrectinib for brain tumors. Trial-level CNS response rate data in NTRK+ brain tumors are limited but consistent with larotrectinib.

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9. Larotrectinib (Vitrakvi) — NTRK inhibitor

Aliases: larotrectinib, Vitrakvi, LOXO-101

Modality: small_molecule · **Regulatory:** FDA-approved (November 2018 accelerated, April 2025 full approval) as a tumor-agnostic indication for solid tumors with NTRK gene fusion without known acquired resistance mutation, metastatic or surgery-likely-to-cause-severe-morbidity, with no satisfactory alternatives or post-progression. Approved by EMA / UK / Japan with similar wording. · **Guidelines:** NCCN tumor-agnostic indication recognized when NTRK fusion is documented. Tumor-agnostic guideline coverage is well-established. · **Geographic scope:** US, EU, UK, Japan all carry tumor-agnostic approval. · **Verified:** 2026-05-26

Larotrectinib is on-label for any NTRK fusion-positive solid tumor including primary GBM — but NTRK fusions in adult GBM are extremely rare (<1%), and the patient has not had NTRK testing yet. As with dabrafenib + trametinib, the biomarker is the load-bearing question.

Next steps

1. Order RNA-based fusion NGS to look for NTRK1/2/3 fusions — this is a known diagnostic gap from profile.json.
2. If NTRK fusion is detected, the prescription path is straightforward; Bayer medical information (1-888-842-2937) provides the off-label literature package and patient support enrollment.
3. If NTRK fusion is negative, drop this row from the dossier. The rate of detection in adult IDH-wildtype GBM is low enough that this is mostly an exclusion test, not a positive-finding test.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT02122913	1/2	completed	unconfirmed	—

Manufacturer / sponsor contact

- **Company:** Bayer (Loxo Oncology subsidiary)
- **Med info phone:**1-888-842-2937
- **Product info:**https://www.vitrakvi-us.com/
- **Compassionate / expanded access:** https://pharma.bayer.com/medical-information-and-product-supply/expanded-access-policy-medicines
- **Notes:** Bayer medical information: 1-888-84-BAYER (1-888-842-2937). Bayer's expanded-access policy is public; case-by-case evaluation.

Payer / coverage: Once NTRK fusion is documented by an FDA-approved test, on-label prescribing is well-supported. Prior auth requires the molecular report. The NCCN Compendium and tumor-agnostic FDA approval clear the payer path.

Notes: April 2025 full approval converted the original accelerated approval. Entrectinib (Rozlytrek) is the alternative TRK inhibitor — entrectinib also covers ROS1 and has broader CNS pharmacology data. Choice of larotrectinib vs entrectinib turns on tolerability and CNS penetration.

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10. Nivolumab (Opdivo)

Aliases: nivolumab, Opdivo, BMS-936558, MDX-1106

Modality: monoclonal_antibody · **Regulatory:** FDA-approved for multiple solid tumors (melanoma, NSCLC, RCC, classical Hodgkin lymphoma, urothelial, etc.) and for MSI-H/dMMR colorectal cancer (tumor-agnostic in CRC, not broader). NOT approved for GBM. CheckMate-143, -498, -548 all failed to show OS benefit in GBM. · **Guidelines:** Not in NCCN for GBM. Tumor-agnostic indications narrower than pembrolizumab's. · **Geographic scope:** US, EU, etc. all approved for multiple non-GBM indications. · **Verified:** 2026-05-26

Nivolumab has a worse GBM story than pembrolizumab — three negative Phase 3 trials (CheckMate-143 in recurrent, -498 in newly diagnosed unmethylated, -548 in methylated) and no tumor-agnostic GBM-relevant biomarker label. Off-label use isn't realistic; if checkpoint blockade is going to be tried in GBM, pembrolizumab via the MSI-H/TMB-H pathways is the better-supported route.

Next steps

1. Do not pursue off-label nivolumab in GBM — the failed trials are too well-known to overcome at payer review.
2. If checkpoint inhibitor access becomes important (e.g. for a hypermutated post-TMZ tumor at recurrence), redirect to pembrolizumab via the MSI-H/TMB-H tumor-agnostic label.
3. Watch the NRG-BN007 readout (NCT04396860) — if it surprises positive, ipi+nivo+RT could become an option for newly diagnosed MGMT-unmethylated patients.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT02617589	3	completed	no	—
NCT02667587	3	completed	no	—
NCT04396860	2/3	completed	no	—

Manufacturer / sponsor contact

- **Company:** Bristol Myers Squibb
- **Med info phone:**1-800-721-5072
- **Product info:**https://packageinserts.bms.com/pi/pi_opdivo.pdf
- **Compassionate / expanded access:** https://www.bms.com/researchers-and-partners/independent-research/early-and-expanded-access-program.html
- **Notes:** BMS Access Support: 1-800-861-0048 (benefits investigation). OPDIVO general info: 1-855-673-4861. BMS Patient Assistance Foundation: 800-736-0003.

Payer / coverage: Off-label GBM coverage is essentially blocked outside MSI-H colorectal — the three failed Phase 3 GBM trials are well-known to payers and used in denials. No defensible off-label path absent a tumor-agnostic biomarker.

Notes: Three negative Phase 3 trials in GBM is unusually strong negative evidence. The NRG-BN007 ipi+nivo combination is the last open question for this class in newly diagnosed unmethylated GBM.

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11. Palbociclib (Ibrance) — CDK4/6 inhibitor

Aliases: palbociclib, Ibrance, PD-0332991

Modality: small_molecule · **Regulatory:** FDA-approved (2015) for HR+/HER2- advanced/metastatic breast cancer in combination with endocrine therapy. No GBM indication. · **Guidelines:** NCCN category 1 for breast cancer. Off-guideline for GBM and not in NCCN Compendium for that use. · **Geographic scope:** Globally available on-label for breast cancer. · **Verified:** 2026-05-26

Palbociclib is FDA-approved for breast cancer and could be prescribed off-label for the patient's CDKN2A-loss biology — but the CNS penetration is poor (Parrish 2015 documented active efflux at the BBB), and the Taal 2018 recurrent-GBM trial was negative. Abemaciclib is the better-positioned CDK4/6 inhibitor for GBM. Palbociclib stays in the dossier as a class member, not as a recommended access target for this patient.

Next steps

1. If a CDK4/6 inhibitor is going to be used off-label for this patient, prefer abemaciclib over palbociclib because of CNS penetration and modestly better preclinical / phase 0 GBM data.
2. If palbociclib is the only option (e.g. abemaciclib intolerance), document the CDKN2A loss and the published rationale, and expect payer pushback.
3. Pfizer medical information (1-800-438-1985) will provide the available off-label palbociclib-in-GBM literature package on request.

Manufacturer / sponsor contact

- **Company:** Pfizer Oncology
- **Med info phone:**1-800-438-1985
- **Product info:**https://www.pfizermedical.com/ibrance
- **Compassionate / expanded access:**https://www.pfizer.com/science/clinical-trials/expanded-access
- **Notes:** Pfizer Oncology Together patient support: 1-877-744-5675. The 1-800-438-1985 line handles both medical information and adverse-event reporting. Pfizer does not operate a published palbociclib expanded-access protocol for GBM.

Payer / coverage: Off-label GBM coverage is a fight — same NCCN Compendium gap as abemaciclib. The Taal recurrent-GBM trial was negative, which weakens the medical-exception argument for palbociclib specifically. Most payers will deny without strong CDKN2A-loss documentation and a citation to the abemaciclib-favoring CNS penetration literature.

Notes: Palbociclib's BBB efflux problem (P-gp / BCRP substrate) is the load-bearing pharmacology issue. The negative Taal 2018 recurrent-GBM trial closed the door on palbociclib monotherapy. Mechanistic rationale survives but is class-level, not molecule-specific.

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12. Pembrolizumab (Keytruda)

Aliases: pembrolizumab, Keytruda, MK-3475

Modality: monoclonal_antibody · **Regulatory:** FDA-approved with multiple tumor-agnostic indications relevant here: (1) MSI-H or dMMR unresectable/metastatic solid tumors post-treatment (full approval Mar 2023); (2) TMB-H (>=10 mut/Mb) unresectable/metastatic solid tumors post-treatment (accelerated approval 2020). EMA-approved for MSI-H/dMMR in five tumor types. · **Guidelines:** NCCN tumor-agnostic recognition for MSI-H/dMMR. The TMB-H accelerated indication is recognized but with more nuance (cutoff debate). NOT a GBM-specific guideline indication outside the biomarker gate. · **Geographic scope:** US, EU, UK, AU, Japan all carry tumor-agnostic MSI-H/dMMR approval. · **Verified:** 2026-05-26

Pembrolizumab is potentially accessible via the tumor-agnostic MSI-H/dMMR or TMB-H pathways — but both biomarkers are currently unknown in this patient and primary MMR-deficient GBM is uncommon. Without that biomarker hit, pembrolizumab monotherapy has failed in three Phase 3 GBM trials and isn't a defensible off-label use. The interesting trial path (NCT07385846 FUS + pembrolizumab) is recurrent-only and MMR-deficient-only, so it's at least one progression and one biomarker hit away.

Next steps

1. Order MMR IHC (MLH1/MSH2/MSH6/PMS2) and TMB by NGS on the resection block — these are listed diagnostic gaps in profile.json.
2. If MSI-H or TMB-H >=10 mut/Mb is found, the tumor-agnostic Keytruda label opens up — call Merck Access Program (855-257-3932) for benefits investigation.
3. If both biomarkers are negative, drop pembrolizumab from the front-line conversation. Three Phase 3 trials in unselected GBM (CheckMate-498, -548, -143) all failed.
4. If the patient later progresses AND turns out to be MMR-deficient, the University of Cincinnati FUS + pembrolizumab trial (NCT07385846) becomes a candidate — call 513-584-7698 once it opens (planned June 2026).

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT02628067	2	completed	unconfirmed	—
NCT07385846	1	not yet recruiting	no	UCCC Clinical Trials Office; cancer@uchealth.com; 513-584-7698

Manufacturer / sponsor contact

- **Company:** Merck Sharp & Dohme
- **Med info phone:**1-800-994-2111
- **Med info email:**merck.medinfo@merck.com
- **Product info:**https://www.merck.com/product/usa/pi_circulars/k/keytruda/keytruda_pi.pdf
- **Compassionate / expanded access:** https://www.merck.com/clinical-trials/clinical-trials-disclosure/individual-patient-expanded-access.html
- **Notes:** Healthcare professional medical information: 1-800-994-2111. Merck Access Program for KEYTRUDA: 855-257-3932 (benefits investigation, prior-auth support, copay assistance). Patient Assistance Program: 855-398-7832 for the KEY+YOU program. Adverse-event reporting: 800-444-2080.

Payer / coverage: Off-label use in GBM (outside MSI-H / TMB-H) faces strong payer pushback — the CheckMate-498 / -548 / -143 series and the Cloughesy 2019 neoadjuvant pembro studies all suggest immune checkpoint monotherapy doesn't work in unselected GBM. Coverage opens up if MSI-H or TMB-H >=10 mut/Mb is documented by an FDA-approved test. Without the biomarker, off-label prescribing is essentially blocked.

Notes: Pembrolizumab's GBM story is a series of negative Phase 3 trials in unselected populations. The biomarker-gated path (MSI-H, TMB-H) is the only off-label justification that holds up under payer review. The Cloughesy 2019 neoadjuvant pembro paper suggested some immune contexture changes but did not translate to OS in larger studies.

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13. Ribociclib (Kisqali) — CDK4/6 inhibitor

Aliases: ribociclib, Kisqali, LEE011

Modality: small_molecule · **Regulatory:** FDA-approved (2017) for HR+/HER2- advanced/metastatic breast cancer and (2024 expanded) for adjuvant HR+/HER2- early breast cancer in combination with endocrine therapy. No GBM indication. · **Guidelines:** NCCN category 1 for breast cancer. Off-guideline for GBM. · **Geographic scope:** Globally available on-label for breast cancer. · **Verified:** 2026-05-26

Ribociclib's published CNS evidence in GBM is limited to the Tien 2019 phase 0 PK/PD study at NYU that showed brain-tumor drug penetration. There has been no subsequent efficacy trial in adult GBM. Off-label use is possible but mechanistically equivalent to abemaciclib without the abemaciclib clinical-development support. Most centers default to abemaciclib if a CDK4/6 inhibitor is going to be tried off-label.

Next steps

1. If a CDK4/6 inhibitor is pursued off-label, prefer abemaciclib first (better GBM-specific clinical data). Ribociclib is the second choice.
2. Call Novartis medical information (1-888-669-6682) for the published Kisqali CNS / phase 0 PK literature package.
3. Novartis Managed Access Program reviews compassionate-use requests case-by-case; not a standing GBM protocol.

Manufacturer / sponsor contact

- **Company:** Novartis
- **Med info phone:**1-888-669-6682
- **Product info:**https://www.us.kisqali.com/
- **Compassionate / expanded access:** https://www.novartis.com/research-development/clinical-trials/managed-access-programs
- **Notes:** Novartis Patient Assistance / medical-information line: 1-888-NOW-NOVA. Novartis has a Managed Access Program portal for compassionate-use requests case-by-case. There is no published Kisqali GBM expanded-access protocol.

Payer / coverage: Same off-label payer pattern as abemaciclib and palbociclib — NCCN Compendium does not list GBM, so off-label prescription faces denial without CDKN2A-loss documentation and rationale letter.

Notes: Tien 2019 phase 0 trial established CNS exposure; downstream efficacy work has not happened.

Ribociclib remains the least-developed CDK4/6 inhibitor in GBM.

Clinical trial (21)

14. B7-H3 CAR T (class — pediatric DFCI + adult China protocols)

Aliases: B7-H3 CAR-T, B7-H3.CD28Z CAR T

Modality: cell_therapy · **Regulatory:** Investigational. Academic INDs across multiple sponsors. · **Guidelines:** Not in NCCN. · **Geographic scope:** US: pediatric only. China: adult but Beijing/Shenzhen only. · **Verified:** 2026-05-26

B7-H3 CAR T is a promising emerging target but the active US trial (DFCI Majzner) is pediatric only (ages 2-21), and the adult-GBM B7-H3 CAR T trial is China-only. Neither is accessible to this US-based 66-year-old. Worth re-screening if an adult US trial opens — Majzner's group has been signaling intent to expand to adults.

Next steps

1. File B7-H3 CAR T as a forward-looking option not currently accessible.
2. Monitor DFCI / Majzner for an adult-cohort expansion; the contact is Susan Chi at 617-632-3270 — call mid-2026 to ask about adult protocol timing.
3. Do not pursue the Zhejiang University NCT04385173 unless the patient is willing to travel to China and absorb the regulatory complexity of receiving an investigational cellular therapy outside the US.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT07390539	I/Ib	not yet recruiting	no	Susan Chi, MD; Susan_Chi@dfci.harvard.edu; 617-632-3270
NCT04385173	I	recruiting	no	Jianmin Zhang, MD; 2307010@zju.edu.cn

Manufacturer / sponsor contact

- **Company:** Multiple academic sponsors (DFCI pediatric program; Zhejiang University adult program)
- **Notes:** No single commercial contact. DFCI pediatric program is age-excluded for this patient; the Chinese adult program is geography-excluded.

Payer / coverage: Investigational.

Notes: B7-H3 is broadly expressed on GBM and across pediatric brain tumors, which is why the pediatric trial came first. The adult-protocol expansion is a known industry signal — Majzner has spoken publicly about wanting to extend to adult HGG.

15. BGB-58067 (brain-penetrant MTA-cooperative PRMT5 inhibitor)

Aliases: BGB-58067, BGB58067

Modality: small_molecule · **Regulatory:** IND-cleared; no approval. Investigational only. · **Guidelines:** Not in any guideline; phase 0/1/2 stage. · **Geographic scope:** US (Phoenix, AZ for the GBM-specific trial); the broader FIH trial NCT06589596 is multi-country. · **Verified:** 2026-05-26

BGB-58067 is the cleanest mechanistic match in the dossier for this patient's MTAP-loss phenotype, the only PRMT5 inhibitor trial currently enrolling NEWLY DIAGNOSED GBM, and it stratifies on MGMT methylation — which is exactly the branching decision the patient needs to make anyway. The catch: single-site in Phoenix and it requires a willingness to do a phase 0 re-resection. If she can travel and is open to a second surgery, this is the top access target.

Next steps

1. Email research@ivybraintumorcenter.org or call the Phase 0 Navigator at 602-406-8605 to ask about MTAP-IHC slot availability and current screening capacity in both Arm A and Arm B.
2. Resolve MGMT methylation status before the screening call — Arm A vs Arm B depends on it.
3. Confirm the patient can travel to Phoenix for screening, the phase 0 re-resection, and follow-up imaging.
4. If Phoenix isn't feasible, ask BeOne Medicines medical information (1-833-969-2463) whether the multi-center NCT06589596 has a GBM-eligible US site closer to the patient.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT07485049	early phase 1	recruiting	likely	Phase 0 Navigator; research@ivybraintumorcenter.org; 602-406-8605
NCT06589596	1	recruiting	unconfirmed	BeOne Medicines Clinical Trial Information Desk; clinicaltrials@beonemed.com

Manufacturer / sponsor contact

- **Company:** BeOne Medicines (formerly BeiGene)
- **Med info phone:**1-833-969-2463
- **Med info email:**medicalinformation@beonemed.com
- **Product info:**https://www.beonemedinfo.com/
- **Notes:** BeOne Medicines US medical information line. The Phoenix trial is technically investigator-sponsored (Sanai) with BeOne as drug supplier — the practical contact for screening is the Ivy navigator, not BeOne.

Payer / coverage: Trial provides drug; patient assumes travel/lodging cost. Ivy Brain Tumor Center sometimes has limited travel-assistance funds — ask the navigator directly.

Notes: BGB-58067 is distinct from anvumetostat (AMG 193) even though both are MTA-cooperative PRMT5 inhibitors — they are different molecules from different sponsors. The trials file has a misleading alias collision; clarified here.

16. CAN-3110 (rQNestin34.5v.2) — replication-competent oncolytic HSV-1

Aliases: CAN-3110, rQNestin34.5v.2

Modality: oncolytic_virus · **Regulatory:** Investigational; FDA Orphan Drug Designation for recurrent high-grade glioma; FDA Fast Track Designation for recurrent HGG. Candel Therapeutics. · **Guidelines:** Not in NCCN. · **Geographic scope:** US — single site at Brigham and Women's / Dana-Farber, Boston. · **Verified:** 2026-05-26

CAN-3110 is the Candel oncolytic HSV-1 with the most encouraging recent GBM data (Ling 2023 Nature, Chiocca interim updates 2025). For THIS patient, the recurrent-only eligibility means CAN-3110 is at least one progression event away. Worth holding in the post-progression plan but not actionable today.

Next steps

1. File CAN-3110 in the post-progression plan, not the front-line plan. Patient must have documented progression on Stupp before she becomes eligible.
2. When the time comes, email info@candeltx.com or contact Chiocca's group at Brigham and Women's neurosurgery directly to ask about current trial slot availability.
3. Confirm GBM tumor is in a non-eloquent area >=1.0-2.0 cm — the protocol requires safe intratumoral injection during stereotactic biopsy.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT03152318		active not recruiting	no	E. Antonio Chiocca, MD PhD (PI, Brigham and Women's / DFCI)

Manufacturer / sponsor contact

- **Company:** Candel Therapeutics
- **Med info email:**info@candeltx.com
- **Product info:**https://candeltx.com/
- **Notes:** Candel does not publish a dedicated medical-information phone line. General company contact info@candeltx.com is the published route. Encouraging interim phase 1b data published in Science Translational Medicine October 2025.

Payer / coverage: Investigational; trial provides drug.

Notes: Nestin-targeted conditional replication is the key biology — the virus replicates preferentially in nestin-positive glioma cells. Cyclophosphamide pretreatment improves intratumoral viral replication.

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17. CARv3-TEAM-E (MGH — CAR T secreting EGFR x CD3 bispecific T-cell engager)

Aliases: CARv3-TEAM-E, INCIPIENT, MGH TEAM construct

Modality: cell_therapy · **Regulatory:** Investigational. Academic IND, Massachusetts General Hospital. · **Guidelines:** Not in NCCN. · **Geographic scope:** US — MGH (Boston) only. · **Verified:** 2026-05-26

MGH's CARv3-TEAM-E is the only EGFRvIII CAR T currently enrolling NEWLY DIAGNOSED patients in the US — Arm 2 specifically targets the MGMT-unmethylated + EGFRvIII+ population. This is the highest-stakes biomarker-gated trial for newly diagnosed disease in the dossier. If the patient turns out to be MGMT-unmethylated AND EGFRvIII+, this is a top-priority option; if either is negative, she's not eligible for Arm 2 and would have to wait for progression to be considered for Arms 1 or 3.

Next steps

1. Order MGMT methylation + EGFRvIII testing on the resection block in parallel — these are the two load-bearing biomarkers for INCIPIENT Arm 2 eligibility.
2. If unmethylated + EGFRvIII+, email carteamingbm@mgb.org urgently — newly diagnosed slots are time-sensitive (cellular product manufacturing takes weeks and ideally starts before RT begins).
3. If unmethylated but EGFRvIII-, the patient might still qualify for Arm 3 (EGFR-amplified) at recurrence — check EGFR amplification status simultaneously.
4. Confirm patient can have an Ommaya reservoir placed (this requires a separate neurosurgical procedure and may be combined with re-resection if needed).

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT05660369		recruiting	unconfirmed	William Curry, MD; carteamingbm@mgb.org; 617-724-6226

Manufacturer / sponsor contact

- **Company:** Massachusetts General Hospital (academic IND)
- **Med info phone:**617-724-6226
- **Med info email:**carteamingbm@mgb.org
- **Product info:**https://www.massgeneral.org/cancer-center/
- **Notes:** MGH academic IND. The CARv3-TEAM-E construct is from the Choi 2024 NEJM paper — first patient durable radiographic response in newly diagnosed unmethylated GBM.

Payer / coverage: Investigational; trial provides product.

Notes: Choi 2024 NEJM published the first patient response — durable radiographic response in MGMT-unmethylated newly diagnosed GBM. The TEAM (T-cell engaging antibody molecule) approach lets the CAR T cell secrete a bispecific that brings local T cells against EGFR — recruits bystander immunity beyond the engineered cell.

18. DCVax-L — autologous dendritic-cell vaccine pulsed with tumor lysate

Aliases: DCVax-L, DCVax, autologous tumor lysate dendritic cell vaccine

Modality: cellular_vaccine · **Regulatory:** Investigational in the US (no BLA filed). UK MHRA Marketing Authorization Application submitted December 2023 under 150-day fast-track review for unmet medical needs; decision expected late 2025 / early 2026. EMA not yet filed. · **Guidelines:** Not in NCCN. · **Geographic scope:**

US: investigational only. UK: pending MHRA review. EU: not filed. Even if UK approves, patient travel + cash-pay is the only realistic international access path. · **Verified:** 2026-05-26

DCVax-L is one of the most-asked-about GBM options because of the Liau 2023 JAMA Oncology external-control analysis showing OS benefit. The hard reality: it's not FDA-approved, the US expanded-access protocol only takes patients whose vaccine was already manufactured under the original phase 3 (i.e. patients enrolled years ago), and new patients post-resection cannot access it through that pathway. The patient's window to get into the manufacturing pipeline at initial resection is already closed. UK MHRA approval is pending; if granted, the access calculus may change.

Next steps

1. Email marnix@nwbio.com to confirm there is no new-patient enrollment route through the existing US protocols. Do NOT raise the patient's hopes about DCVax-L until that confirmation is in hand.
2. Monitor UK MHRA decision (expected late 2025 / early 2026). If approved, the question of whether US patients can access via international cash-pay reopens.
3. If the family has unlimited resources and is determined to pursue DCVax-L, the realistic path is to follow the UK approval and then ask Northwest Biotherapeutics about international compassionate-use options post-approval.
4. In the meantime, the patient-actionable autologous-cell vaccine path is the SurVaxM trial (NCT05163080) at Roswell Park or partner US sites — see the SurVaxM access row.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT00045968	3	active not recruiting	no	—
NCT02146066	expanded access	available	no	Marnix Bosch, MBA PhD; marnix@nwbio.com; 240-497-9022

Manufacturer / sponsor contact

- **Company:** Northwest Biotherapeutics
- **Med info phone:**240-497-9022
- **Med info email:**marnix@nwbio.com
- **Product info:**https://nwbio.com/
- **Compassionate / expanded access:**https://clinicaltrials.gov/study/NCT02146066
- **Notes:** Northwest Biotherapeutics CMO Marnix Bosch is the published EAP contact. Cost of treatment historically estimated at $200-300K per patient (manufacturing-cost limited). Production now scaling on the Flaskworks EDEN automated platform.

Payer / coverage: No US payer coverage — DCVax-L has no FDA approval, no BLA filing, and no Medicare coverage. Expanded-access participants historically pay out-of-pocket or self-fund through fundraising. UK MHRA decision is the next regulatory inflection. If approved in the UK, US patients could in principle access it via international travel + cash pay, but the manufacturing footprint is currently the bottleneck.

Notes: The Liau 2023 JAMA Oncology publication used external-control comparators which is the methodologically controversial part of the regulatory submission. NWBO has no announced US BLA timeline. The cash-pay cost of historical DCVax-L production was ~$200-300K per patient.

19. DNX-2401 (tasadenoturev / Delta-24-RGD) — oncolytic adenovirus

Aliases: DNX-2401, tasadenoturev, Delta-24-RGD

Modality: oncolytic_virus · **Regulatory:** Investigational. DNAtrix (Houston, TX). FDA Orphan Drug Designation for HGG. · **Guidelines:** Not in NCCN. · **Geographic scope:** US (MD Anderson historically the lead site). · **Verified:** 2026-05-26

DNX-2401 + pembrolizumab generated interesting Phase 2 data (Lang 2023 Nat Med) but did not formally hit its prespecified efficacy threshold. There is currently no actively-recruiting US trial of DNX-2401 — the CAPTIVE study is closed and no successor trial is announced. For this patient, DNX-2401 is mechanistic evidence rather than a near-term access path.

Next steps

1. Treat DNX-2401 as forward-looking — call DNAtrix general inquiry (via the dnatrix.com website) to ask whether a successor trial is in planning.
2. If oncolytic-virus access is the goal, redirect to CAN-3110 (Brigham/DFCI) or PVSRIPO (Duke) — both have currently-open US trials, both recurrent-only.

Manufacturer / sponsor contact

- **Company:** DNAtrix, Inc.
- **Product info:**https://www.dnatrix.com/
- **Notes:** DNAtrix continues to develop DNX-2401 in combination with pembrolizumab (CAPTIVE / KEYNOTE-192). The recent publication (Lang 2023 Nat Med) reported 10.4% ORR + 52.7% 12-month OS, missing the prespecified statistical threshold but showing durable responses in a subset. The development path forward depends on whether DNAtrix can identify a responder-enriched subgroup.

Payer / coverage: Investigational; trial provides drug if a new study opens.

Notes: Foundational evidence: Lang 2017 J Clin Oncol (Phase 1) and Lang 2023 Nat Med (Phase 2). The Delta-24-RGD construct deletes E1A and adds RGD-4C integrin-binding for retargeting away from CAR receptor.

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20. Dual-targeting CAR-NK cells (Beijing Biotech)

Aliases: Dual-targeting CAR-NK, Beijing Biotech CAR-NK

Modality: cell_therapy · **Regulatory:** Investigational. China-only IND. · **Guidelines:** Not in NCCN. · **Geographic scope:** China only (Peking University Shenzhen Hospital). · **Verified:** 2026-05-26

Dual-targeting CAR-NK is geography-bounded to China and recurrent-only. Not realistically accessible to a US-based patient at this stage in her disease course. Listed for dossier completeness.

Next steps

1. Do not pursue — geography mismatch.
2. If the user wants to explore CAR-NK biology, redirect to academic US programs (limited GBM-specific options currently).

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT07480941	1	recruiting	no	Seni S Lu, PhD; Seni-Lu@beijing-biotech.com; +86 13076790030

Manufacturer / sponsor contact

- **Company:** Beijing Biotech
- **Med info email:**Seni-Lu@beijing-biotech.com
- **Notes:** China-based development-stage biotech. No US presence.

Payer / coverage: Not applicable for US patients.

Notes: CAR-NK has lower CRS / ICANS profile than CAR-T in theory — but the GBM-specific clinical data are very early. The interesting US-comparable programs are still preclinical.

21. E-SYNC — UCSF anti-EGFRvIII synNotch -> anti-EphA2/IL-13Ra2 CAR T cells

Aliases: E-SYNC, synNotch CAR-T, anti-EGFRvIII synNotch

Modality: cell_therapy · **Regulatory:** Investigational. Academic IND, UCSF. · **Guidelines:** Not in NCCN. · **Geographic scope:** US — UCSF (San Francisco) only. · **Verified:** 2026-05-26

UCSF's E-SYNC is the synNotch-architecture CAR T — the construct uses EGFRvIII as a recognition trigger that then induces expression of a second CAR against EphA2 / IL-13Ra2 (broader expression on glioma cells). Open and recruiting for both newly diagnosed AND recurrent EGFRvIII+ GBM, which makes it more flexible than the Penn / MGH options. Same biomarker gate as the rest of this class — confirm EGFRvIII first.

Next steps

1. Order EGFRvIII testing — diagnostic gap from profile.json.
2. If EGFRvIII positive AND H-score is high enough for Cohort 2 (or any-positive for Cohort 1), call UCSF Neuro-Oncology New Patient Coordinator at 877-827-3222.
3. Newly diagnosed enrollment is time-sensitive — cellular product manufacturing typically takes 4-6 weeks; start the conversation as soon as the biomarker is confirmed.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT06186401	1	recruiting	unconfirmed	Neuro-Oncology New Patient Coordinator; 877-827-3222

Manufacturer / sponsor contact

- **Company:** UCSF (academic IND)
- **Med info phone:**877-827-3222
- **Med info email:**NeuroOncNewPatientCoord@ucsf.edu
- **Product info:**https://www.ucsfhealth.org/conditions/glioblastoma/treatment
- **Notes:** UCSF Brain Tumor Center is the manufacturing and clinical home.

Payer / coverage: Investigational; trial provides product.

Notes: Hideho Okada's lab has been the leading academic developer of synNotch CAR T approaches for GBM. The biology elegantly addresses antigen heterogeneity — EGFRvIII triggers, then the cell goes after EphA2/IL-13Ra2 which are more uniformly expressed.

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22. EGFRvIII-directed CAR T cells (class — multiple constructs)

Aliases: EGFRvIII CAR-T, anti-EGFRvIII CAR, humanized EGFRvIII CAR, Penn O'Rourke construct, NCI Clone 139

Modality: cell_therapy · **Regulatory:** Investigational. Multiple academic INDs across Penn, NCI, MGH, Duke, UCSF. No FDA approval. · **Guidelines:** Not in NCCN. · **Geographic scope:** US: multiple academic centers (Penn, UCSF, MGH, Duke). China: Beijing-only protocols geography-bounded. · **Verified:** 2026-05-26

EGFRvIII CAR-T is the class of cellular therapies most-studied in GBM. The patient's EGFR amplification / EGFRvIII status is currently unknown — that's the biomarker gate before any of these trials becomes relevant. AND all currently open US protocols are recurrent-only (Penn 215-349-8245, UCSF, MGH INCIPIENT) — newly-diagnosed enrollment is rare in this class. So even if she's EGFRvIII+, the access window opens at first progression, not at diagnosis.

Next steps

1. Order EGFRvIII testing (RT-PCR or NGS fusion call) on the resection block — this is a known diagnostic gap from profile.json.
2. Order EGFR amplification testing by FISH or NGS copy-number — separate but related call.
3. File EGFRvIII CAR-T programs in the post-progression plan, conditional on the biomarker result.
4. If EGFRvIII is positive, individual program contacts: Penn (215-349-8245), UCSF (877-827-3222 for E-SYNC), MGH (617-724-6226 for CARv3-TEAM-E).

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT07209241		recruiting	no	Abramson Cancer Center Clinical Trials; 215-349-8245
NCT05802693		not yet recruiting	no	Yang Xuejun; yxja03728@btch.edu.cn

Manufacturer / sponsor contact

- **Company:** Multiple academic sponsors (Penn / Marcela Maus at MGH / Duke / UCSF / Bagley at Penn) + industry collaborators
- **Notes:** EGFRvIII CAR-T is an academic-IND landscape, not a commercial product. No single corporate contact; each protocol has its own center contact (see individual trial rows: Penn 215-349-8245; UCSF E-SYNC 877-827-3222; MGH INCIPIENT 617-724-6226).

Payer / coverage: Investigational; trial provides product.

Notes: Foundational evidence: O'Rourke 2017 STM (the first published EGFRvIII CAR-T trial); Goff 2019 JIT (NCI construct); Choi 2024 NEJM (MGH CARv3-TEAM-E intraventricular); Bagley 2024 Nat Med (Penn bivalent intrathecal). The class is rapidly evolving — new constructs (synNotch, TEAM, dual-CAR) every 12-18 months.

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23. Focused ultrasound BBB opening (SonoCloud-9 / NaviFUS / Sonabend FUS programs)

Aliases: SonoCloud-9, Carthera, NaviFUS, focused ultrasound BBB opening, FUS-BBB

Modality: device · **Regulatory:** Investigational. FDA Breakthrough Device Designation for SonoCloud-9 (2022). FDA / EMA Orphan Drug Designation 2023 for carboplatin + SonoCloud. No US device approval. · **Guidelines:** Not in NCCN. · **Geographic scope:** Multi-site US + Europe. US sites include Northwestern, Columbia, MD Anderson and others. · **Verified:** 2026-05-26

Focused ultrasound BBB opening is the most clinically advanced strategy to improve drug delivery to glioblastoma — but every active trial is RECURRENT-only and most require pre-planned re-resection (Carthera SONOBIRD) or biomarker eligibility (NaviFUS + MMR-deficient pembro at UC). For this newly-diagnosed patient, FUS-BBB is at least one progression away. If she progresses and her US site has SONOBIRD or a partner protocol open, that becomes a serious candidate at first recurrence.

Next steps

1. File FUS-BBB programs in the post-progression plan.
2. When the time comes, the access strategy is: (a) confirm patient is a surgical candidate for re-resection, (b) call the nearest SONOBIRD US site (Northwestern via Sonabend lab, Columbia, MD Anderson, etc.), or (c) if MMR-deficient by then, also explore NCT07385846 at UC.
3. The Carthera SonoCloud-9 implant is placed at the first recurrence resection — the patient must be a surgical candidate AND aware that the device stays in place across multiple carboplatin infusions.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT04528680	3	recruiting	no	—
NCT07385846	1	not yet recruiting	no	UCCC Clinical Trials Office; cancer@uchealth.com; 513-584-7698

Manufacturer / sponsor contact

- **Company:** Carthera (SonoCloud-9); NaviFUS Corporation (navigated FUS)

- **Product info:**https://carthera.eu/
- **Notes:** Carthera is the lead device sponsor for SonoCloud-9. NaviFUS is the alternative external-applicator device. Both are investigational in the US; no commercial medical-information line.

Payer / coverage: Investigational; trials provide device. Carboplatin is a SoC drug — covered under standard chemo benefits.

Notes: Sonabend 2023 Lancet Oncology phase 1/2 established proof-of-concept for SonoCloud-9 + albumin-bound paclitaxel in recurrent GBM. SONOBIRD is the registrational phase 3 with IV carboplatin as the partner drug. NaviFUS is the alternative external-applicator approach without an implant.

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24. GBM AGILE — adaptive platform trial (multi-arm)

Aliases: GBM AGILE, Global Adaptive Trial Master Protocol, VT1021, troriluzole, ADI-PEG 20, AZD1390, tinostamustine

Modality: platform_trial · **Regulatory:** Multiple investigational arms; standard-of-care control arm. Platform run by Global Coalition for Adaptive Research (GCAR). · **Guidelines:** Not in NCCN — but the platform structure means experimental arms can graduate to NCCN if they hit efficacy thresholds. · **Geographic scope:** US + international (Europe, Canada, Australia). >50 US sites currently active. · **Verified:** 2026-05-26

GBM AGILE is the single best 'multi-shot' option for the patient who wants to access experimental therapy without locking in a specific drug. It runs at >50 US sites, takes newly diagnosed patients with the standard-of-care backbone built in, and the Bayesian algorithm assigns the patient to whichever experimental arm has the highest current posterior probability of benefit. The catch: she doesn't get to choose the arm. If she is committed to a specific mechanism (e.g. MTAP-directed therapy), this isn't the right access path — the assignment is algorithmic.

Next steps

1. Call GCAR patient info at 310-598-3199 or email patientinfo@gcaresearch.org to identify the nearest GBM AGILE site to the patient.
2. Confirm patient's MGMT status before screening — the platform stratifies by MGMT.
3. Discuss with the patient whether she's comfortable with algorithmic arm assignment vs choosing a specific trial (like BGB-58067) — this is a values question, not a clinical one.
4. If she enrolls in GBM AGILE, she gives up the ability to enroll in BGB-58067 / MountainTAP-5 / SurVaxM simultaneously — the trial is a one-shot commitment.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT03970447	2/3	recruiting	likely	Rachel Rosenstein-Sisson (GCAR); patientinfo@gcaresearch.org; 310-598-3199

Manufacturer / sponsor contact

- **Company:** Global Coalition for Adaptive Research (sponsor); individual drug arms have their own sponsors

(Vigeo Therapeutics for VT1021, Biohaven for troriluzole, Polaris for ADI-PEG 20, AstraZeneca for AZD1390, Mundipharma for tinostamustine)
- **Med info phone:**310-598-3199
- **Med info email:**patientinfo@gcaresearch.org
- **Product info:**https://www.gcaresearch.org/gbmagile
- **Notes:** GCAR is a nonprofit research coalition. The platform is the access point — once screened, the algorithm assigns the patient to an arm. Patient cannot pre-select.

Payer / coverage: Trial provides investigational drugs and standard-of-care backbone; payer covers routine care costs only.

Notes: GBM AGILE is the most well-organized GBM platform trial in the world. The Bayesian adaptive design is good for the population (high-yield experimental-arm selection) but offers less individual control than a single-arm trial. Worth screening AT THE SAME TIME as the targeted-trial conversation so the patient has both options in hand before deciding.

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25. IDE397 — MAT2A inhibitor (IDEAYA)

Aliases: IDE397, IDE-397

Modality: small_molecule · **Regulatory:** IND-cleared; no approval. IDEAYA Biosciences. · **Guidelines:** Not in any guideline. · **Geographic scope:** US — but CNS-excluded. · **Verified:** 2026-05-26

IDE397 is IDEAYA's MAT2A inhibitor — mechanistically interesting for the MTAP-null biology but the only active trial (NCT04794699) explicitly excludes primary CNS malignancy. Not a realistic access path for this patient unless IDEAYA opens a CNS-specific protocol later.

Next steps

1. Do not pursue NCT04794699 enrollment — primary CNS malignancy is an exclusion criterion.
2. If the user wants forward-looking IDEAYA contact, email IDEAYAClinicalTrials@ideayabio.com asking whether any IDE397 CNS-specific trial is in the pipeline.
3. Redirect MAT2A-axis interest to potential combination arms within MountainTAP-5 (BMS uses navlimetostat ± other agents) or to mechanism-only evidence — there is no patient-accessible MAT2A trial for GBM today.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT04794699		active not recruiting	no	IDEAYA Clinical Trials; 1-855-433-2246

Manufacturer / sponsor contact

- **Company:** IDEAYA Biosciences
- **Med info phone:**1-855-433-2246
- **Med info email:**IDEAYAClinicalTrials@ideayabio.com

- **Product info:**https://www.ideayabio.com/
- **Notes:** IDEAYA general phone (650) 443-6209. Clinical-trials desk is the better contact for investigational-drug inquiries.

Payer / coverage: Investigational; not applicable.

Notes: IDEAYA's CDKN2A/MTAP strategy is broader (MAT2A + PRMT5 combinations with IDE892, urothelial expansions with sacituzumab) but the CNS gate is the load-bearing constraint for this case.

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26. IDE892 — next-generation MTA-cooperative PRMT5 inhibitor (IDEAYA)

Aliases: IDE892, IDE-892

Modality: small_molecule · **Regulatory:** IND-cleared 2025; no approval. IDEAYA Biosciences. · **Guidelines:** Not in any guideline. · **Geographic scope:** US — but CNS-excluded. · **Verified:** 2026-05-26

IDE892 is IDEAYA's next-generation MTA-cooperative PRMT5 inhibitor with proposed combination synergy with IDE397 (MAT2A). The trial excludes primary CNS malignancy, so this patient is not enrollable today. Worth re-screening if IDEAYA opens a CNS-specific cohort or amends the protocol.

Next steps

1. Do not pursue NCT07277413 — primary CNS exclusion.
2. Ask IDEAYA Clinical Trials (1-855-433-2246) whether a CNS / GBM cohort is planned for 2026-2027 — the dual PRMT5 + MAT2A mechanism is highly relevant for MTAP-null biology.
3. Treat IDE892 as forward-looking pipeline; the patient-actionable PRMT5 options remain BGB-58067 (Phoenix) and the upcoming MountainTAP-5 (BMS).

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT07277413		recruiting	no	IDEAYA Clinical Trials; 1-855-433-2246

Manufacturer / sponsor contact

- **Company:** IDEAYA Biosciences
- **Med info phone:**1-855-433-2246
- **Med info email:**IDEAYAClinicalTrials@ideayabio.com
- **Product info:**https://www.ideayabio.com/
- **Notes:** Same IDEAYA contact as IDE397.

Payer / coverage: Investigational; not applicable.

Notes: First-patient-in announced 2026. IDE892's MTA-positive / SAM-negative cooperativity profile is theoretically optimized for the MTAP-deleted biology, and the IDE892 + IDE397 combination produced complete and durable responses in MTAP-deleted preclinical models per the AACR 2026 disclosure.

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27. Lerapolturev (PVSRIPO) — recombinant oncolytic poliovirus

Aliases: PVSRIPO, lerapolturev, PVS-RIPO

Modality: oncolytic_virus · **Regulatory:** Investigational. FDA Breakthrough Therapy Designation for recurrent GBM. FDA Orphan Drug Designation for melanoma and GBM. Istari Oncology. · **Guidelines:** Not in NCCN. · **Geographic scope:** US — Duke is the lead site. International sites have not been announced for LUMINOS-101. · **Verified:** 2026-05-26

Lerapolturev / PVSRIPO is the Duke recombinant poliovirus and has the longest GBM clinical history of any oncolytic virus program (Desjardins 2018 NEJM). The currently-recruiting LUMINOS-101 phase 2 is recurrent-only at Duke — newly-diagnosed patients are at least one progression event away. The polio-booster requirement is unusual and needs to be scheduled in advance of any infusion.

Next steps

1. File PVSRIPO in the post-progression plan. Document polio vaccination status now; if the patient hasn't had a recent booster, schedule one — it's a protocol requirement for LUMINOS-101 enrollment.
2. When the time comes, email dukebrain1@dm.duke.edu or call Madison Shoaf at 919-684-5301 for Duke screening.
3. Confirm supratentorial tumor location and KPS >=70 at the time of progression.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT06177964	2	recruiting	no	Madison Shoaf, MD; dukebrain1@dm.duke.edu; 919-684-5301
NCT02986178	1	completed	no	—

Manufacturer / sponsor contact

- **Company:** Istari Oncology
- **Med info email:**MedicalAffairs@istarioncology.com
- **Product info:**https://www.istarioncology.com/
- **Notes:** Istari is development-stage; medical affairs email is the published route. The randomized phase 2 (LUMINOS-101) is the registrational follow-up to the Desjardins 2018 NEJM phase 1.

Payer / coverage: Investigational; trial provides drug.

Notes: PVSRIPO has been administered to >200 patients across multiple trials. The Desjardins 2018 phase 1 showed durable responses in a subset of patients. The cervical perilymphatic injection component is a newer addition based on systemic immune-priming biology (Yang 2023).

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28. M032 — oncolytic HSV-1 expressing IL-12

Aliases: M032, M032-HSV, oHSV-IL12

Modality: oncolytic_virus · **Regulatory:** Investigational. University of Alabama Birmingham program (Friedman / Markert). · **Guidelines:** Not in NCCN. · **Geographic scope:** US — UAB only when active. · **Verified:** 2026-05-26

M032 is the UAB oncolytic HSV-1 with IL-12 transgene — interesting preclinical biology (Roth 2014) and early-phase clinical experience but no actively-recruiting US trial open for adult GBM in the current dossier. For this patient, M032 is forward-looking and not a near-term access path. If interested, call UAB neurosurgery directly to ask about current screening status.

Next steps

1. Call UAB neurosurgery (the published M032 program contacts are Burt Nabors and Renee Chambers) to ask whether any M032 adult GBM trial is currently enrolling.
2. If oncolytic-virus access is the goal, the patient-actionable US options remain CAN-3110 (Brigham) and PVSRIPO (Duke) — both recurrent-only.

Manufacturer / sponsor contact

- **Company:** Treovir LLC / UAB academic IND
- **Notes:** M032 was originally a UAB academic IND; commercial development partner Treovir LLC. The active early-phase trials are run out of UAB neurosurgery (Burt Nabors, Renee Chambers).

Payer / coverage: Investigational.

Notes: M032 expresses IL-12 from within the oncolytic HSV-1 backbone — combines oncolysis with local cytokine delivery. UAB's program is one of the longer-running academic oncolytic HSV efforts.

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29. MRTX1719 / Navlimetostat (BMS-986504) — MTA-cooperative PRMT5 inhibitor

Aliases: MRTX1719, navlimetostat, BMS-986504

Modality: small_molecule · **Regulatory:** IND-cleared; no approval. Bristol Myers Squibb development program (acquired with Mirati in 2023). · **Guidelines:** Not in any guideline; phase 1/2 stage. · **Geographic scope:** US + multi-country. · **Verified:** 2026-05-26

MRTX1719 / navlimetostat is the BMS MTA-cooperative PRMT5 program. The currently-recruiting NCT05245500 phase 1 does not name GBM in its tumor list, so eligibility for this patient is uncertain — call the BMS contact center to confirm. The bigger opportunity is NCT07492680 (MountainTAP-5), which opens July 2026 with an explicit GBM cohort combining the drug with TMZ + RT — that's the BMS successor program and is the most patient-relevant option once it activates.

Next steps

1. Call BMS Clinical Trials Contact Center at 855-907-3286 to confirm whether NCT05245500 will accept a primary brain tumor (GBM) under the existing protocol amendments.
2. Mark MountainTAP-5 (NCT07492680) for re-screening the week it opens (planned July 2026) — its Cohort 4 (BMS-986504 + TMZ + RT) is purpose-built for newly diagnosed MTAP-deleted GBM and is the BMS successor strategy.

3. If the Phoenix BGB-58067 trial isn't feasible, MountainTAP-5 is the second-best PRMT5 option for this patient — both target the same biology but with different molecules and different protocol structures.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT05245500	1	recruiting	unconfirmed	BMS Clinical Trials Contact Center; Clinical.Trials@bms.com; 855-907-3286
NCT07492680	2	not yet recruiting	likely	BMS Clinical Trials Contact Center; Clinical.Trials@bms.com; 855-907-3286

Manufacturer / sponsor contact

- **Company:** Bristol Myers Squibb (acquired Mirati 2023)
- **Med info phone:**1-800-721-5072
- **Med info email:**Clinical.Trials@bms.com
- **Product info:**https://www.bms.com/
- **Compassionate / expanded access:** https://www.bms.com/researchers-and-partners/independent-research/early-and-expanded-access-program.html
- **Notes:** BMS adverse-event / medical-information line for product safety: 1-800-721-5072. BMS Patient Assistance Foundation (independent charity for free product for uninsured patients with approved BMS commercial drugs): 800-736-0003 — not relevant for investigational MRTX1719 but useful for nivolumab/ipilimumab co-enrollment.

Payer / coverage: Investigational; drug provided through trial. No payer issues for the drug itself.

Notes: MRTX1719 originated at Mirati Therapeutics and transferred to BMS via the 2023 acquisition. The development code BMS-986504 and the INN navlimetostat refer to the same molecule. The MountainTAP-5 protocol also includes combination arms with daraxonrasib (RAS(ON)), nivolumab + relatlimab, and chemotherapy — i.e. a basket platform on the MTA-cooperative backbone.

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30. OH2 — oncolytic HSV-2 (Binhui)

Aliases: OH2

Modality: oncolytic_virus · **Regulatory:** Investigational. China-only. · **Guidelines:** Not in NCCN. · **Geographic scope:** China only (Beijing Tiantan Hospital). · **Verified:** 2026-05-26

OH2 is the Binhui oncolytic HSV-2. China-only and recurrent-only — not accessible to this US-based newly-diagnosed patient. Listed for completeness.

Next steps

1. Do not pursue — geography mismatch.
2. If oncolytic HSV is of interest, redirect to CAN-3110 (Candel, Brigham/DFCI) or M032 (UAB) — both US

programs in this class.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT05235074	1/2a	recruiting	no	Wenbin Li; neure55@126.com; +86 15301377998

Manufacturer / sponsor contact

- **Company:** Binhui Biopharmaceutical Co., Ltd.
- **Notes:** China-based. No US presence or compassionate-use program for US patients.

Payer / coverage: Not applicable.

Notes: Foreign-origin oncolytic HSV programs. The US-accessible options in this mechanism class are CAN-3110 (Candel) and M032 (UAB-stage; not currently in the active trial set).

31. Penn CART-EGFR-IL13Ra2 (Bagley bivalent intrathecal CAR T)

Aliases: CART-EGFR-IL13Ra2, Bagley Penn construct, bivalent EGFR-IL13Ra2 CAR

Modality: cell_therapy · **Regulatory:** Investigational. Academic IND, University of Pennsylvania. · **Guidelines:** Not in NCCN. · **Geographic scope:** US — Penn (Philadelphia) only. · **Verified:** 2026-05-26

Penn's bivalent EGFR-IL13Ra2 CAR T is the next-generation construct that builds on Bagley 2024 Nature Medicine. Open and recruiting at Penn, but only for recurrent IDH-wildtype GBM with confirmed wild-type EGFR amplification. The patient is newly diagnosed AND her EGFR amplification status is unknown, so this is two gates away: progression + biomarker confirmation.

Next steps

1. Order EGFR amplification testing on the resection block (FISH or NGS copy-number) — diagnostic gap from profile.json.
2. File Penn CART-EGFR-IL13Ra2 in the post-progression plan, conditional on EGFR amplification positivity.
3. When the time comes, email PMCancerResearch@pennmedicine.upenn.edu or call 215-349-8245 for screening.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT07209241	1	recruiting	no	Abramson Cancer Center Clinical Trials; 215-349-8245

Manufacturer / sponsor contact

- **Company:** University of Pennsylvania (academic IND, no industry partner publicly named)
- **Med info phone:**215-349-8245
- **Med info email:**PMCancerResearch@pennmedicine.upenn.edu
- **Product info:**https://www.pennmedicine.org/cancer/clinical-trials
- **Notes:** Penn Abramson Cancer Center is the manufacturing and clinical home for this construct. Academic IND — no commercial sponsor handles compassionate use.

Payer / coverage: Investigational; trial provides product.

Notes: Bagley 2024 Nat Med published the first-in-human bivalent CART results — durable radiographic responses in a subset, intrathecal administration. The next-generation trial is testing dosing schedules.

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32. S095035 — MAT2A inhibitor (Servier)

Aliases: S095035, S-095035, CL1-95035-001

Modality: small_molecule · **Regulatory:** IND-cleared; no approval. Servier Bio-Innovation. · **Guidelines:** Not in any guideline. · **Geographic scope:** Multi-country trial with explicit China-mainland exclusion for IDHwt GBM. · **Verified:** 2026-05-26

Servier's S095035 is the only MAT2A inhibitor trial that explicitly listed IDH-wildtype GBM as an eligible tumor type — but it's currently active not recruiting, so a new slot would require either a protocol amendment or program continuation. Call Servier US medical information (1-800-770-0631) to ask whether the trial is re-opening or whether an extension cohort is planned.

Next steps

1. Call Servier medical information at 1-800-770-0631 and ask specifically whether NCT06188702 will re-open enrollment for the IDHwt GBM cohort.
2. If the trial is permanently closed to new enrollment, treat S095035 as mechanistic evidence and redirect MAT2A interest to the BMS MountainTAP-5 navlimetostat combinations (which include MAT2A-axis collaborators) or IDEAYA's combination program (CNS-excluded).
3. AG-270 / S095033 (the first-generation Servier MAT2A inhibitor) was used in the published phase 1 trial (Sumi 2024 Nat Commun) but development was discontinued; it is not a current access path.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT06188702		active not recruiting	unconfirmed	—

Manufacturer / sponsor contact

- **Company:** Servier Bio-Innovation LLC (US subsidiary of Servier, France)
- **Med info phone:**1-800-770-0631
- **Product info:**https://servier.us/
- **Notes:** Servier US medical information line. AG-270 / S095033 is the related first-generation MAT2A inhibitor (Marjon/Kalev 2021 Nat Commun paper) which informed the S095035 design.

Payer / coverage: Investigational; not applicable.

Notes: Servier's MAT2A program is the most CNS-friendly MAT2A trial design currently registered. The active-not-recruiting status is the live constraint. Worth a phone call before writing it off.

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33. SurVaxM — survivin peptide vaccine

Aliases: SurVaxM, SVN53-67/M57-KLH

Modality: peptide_vaccine · **Regulatory:** Investigational; no approval. MimiVax (developed at Roswell Park). · **Guidelines:** Not in NCCN. · **Geographic scope:** US only; 11 sites across the SURVIVE trial. · **Verified:** 2026-05-26

SurVaxM is the most patient-relevant autologous-vaccine option for a newly diagnosed US GBM patient — phase 2b is multi-site US, the regimen layers cleanly on top of standard Stupp (vaccination starts after RT + TMZ completion), and the published phase 2 (Ahluwalia 2023 J Clin Oncol) showed promising survival signal. The exclusion of prior immunotherapy and Optune means the patient should decide between SurVaxM and Optune at the maintenance-TMZ branch point. The current 'active not recruiting' status means call before assuming a slot.

Next steps

1. Call Roswell Park (1-800-ROSWELL / 1-800-767-9355) to ask whether the SURVIVE trial has any open screening capacity at any of the 11 US sites — the registry says active not recruiting but site-level reactivations sometimes happen.
2. If a slot is available, the patient must commit BEFORE starting Optune (the trial excludes prior Optune use). This is a forced choice between SurVaxM and TTFields at the maintenance branch.
3. Eudocia Lee at DFCI and the Roswell Park PI Ahluwalia / Kaur teams are the main published investigators; if Roswell is full, DFCI may have a partner-site slot.
4. Confirm MGMT methylation before SurVaxM enrollment — the trial requires the MGMT result to be reported.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT05163080	2b	active not recruiting	likely	AskRoswell@Roswellpark.org; 1-800-767-9355

Manufacturer / sponsor contact

- **Company:** MimiVax, LLC (developed at Roswell Park Comprehensive Cancer Center)
- **Med info phone:**1-800-767-9355
- **Med info email:**AskRoswell@Roswellpark.org
- **Product info:**https://www.mimivax.com/survaxm/
- **Notes:** Roswell Park 1-800-ROSWELL is the practical screening line for the SURVIVE trial. MimiVax does not have a dedicated commercial medical-information desk because the drug is investigational.

Payer / coverage: Investigational; trial provides drug.

Notes: Ahluwalia 2023 J Clin Oncol established the phase 2 survival signal. SURVIVE is the randomized confirmatory phase 2b. The SurVaxM / Optune mutual-exclusion is the practical decision point at the end of concurrent RT + TMZ.

34. TNG462 / Vopimetostat — peripheral MTA-cooperative PRMT5 inhibitor

Aliases: TNG462, vopimetostat

Modality: small_molecule · **Regulatory:** IND-cleared; no approval. Tango Therapeutics. · **Guidelines:** Not in any guideline. · **Geographic scope:** US + multi-country. · **Verified:** 2026-05-26

TNG462 is the Tango peripheral PRMT5 inhibitor — it doesn't penetrate the brain adequately, so it's not the right tool for primary GBM. The Tango brain-penetrant successor TNG456 is the molecule to watch; ask the clinicaltrials@tangotx.com contact whether a TNG456 GBM trial is open or planned. For this patient, the patient-relevant Tango option is TNG456, not TNG462.

Next steps

1. Do not pursue TNG462 enrollment — wrong tumor biology (peripheral compartment).
2. Email clinicaltrials@tangotx.com asking specifically about TNG456 (brain-penetrant successor) trial timing and any planned GBM-specific protocols.
3. Treat Tango's PRMT5 program as a forward-looking option, not a near-term access path for this patient.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT05732831	1/2	recruiting	no	Maxim Pimkin, MD; clinicaltrials@tangotx.com; 857-320-4899

Manufacturer / sponsor contact

- **Company:** Tango Therapeutics
- **Med info phone:**857-320-4899
- **Med info email:**clinicaltrials@tangotx.com
- **Product info:**https://www.tangotx.com/
- **Notes:** Tango is a development-stage biotech — no commercial product, no dedicated medical-information line beyond the clinical-trial contact above. For brain-penetrant successor TNG456 inquiries, use the same contact and ask about TNG456 trial timing.

Payer / coverage: Investigational; trial provides drug.

Notes: Tango shifted focus from TNG908 (terminated for GBM lack-of-benefit) to TNG462 (peripheral, expanded to NSCLC/pancreatic/cholangiocarcinoma) and TNG456 (brain-penetrant, entered clinic 2025). The brain-penetrant Tango program is now TNG456, not TNG462 or TNG908.

Unavailable (6)

35. AG-270 / S095033 — first-generation MAT2A inhibitor

Aliases: AG-270, AG270, S095033, S-095033

Modality: small_molecule · **Regulatory:** Development discontinued. AG-270 was the Agios MAT2A inhibitor (later Servier S095033); the phase 1 study (Sumi 2024 Nat Commun) read out and the program did not advance to phase 2. · **Guidelines:** Not in any guideline. · **Geographic scope:** Globally discontinued. · **Verified:** 2026-05-26

AG-270 is mechanistic evidence only — the first-generation Servier MAT2A inhibitor published its phase 1 readout in Nature Communications (Sumi 2024) and the program did not advance to phase 2 development. The patient-actionable MAT2A path is S095035 (Servier successor) or the MountainTAP-5 BMS combinations.

Next steps

1. Treat AG-270 as foundational evidence only.
2. Pursue MAT2A access through the S095035 row (Servier) or the BMS MountainTAP-5 navlimetostat combination arms.

Manufacturer / sponsor contact

- **Company:** Servier (acquired Agios oncology portfolio 2021)
- **Med info phone:**1-800-770-0631
- **Product info:**https://servier.us/
- **Notes:** Same Servier medical-information line as S095035. AG-270 / S095033 informed but is no longer in active development; the successor is S095035 (still investigational, see separate row).

Payer / coverage: Not applicable.

Notes: Originally Agios. Acquired by Servier when Servier bought the Agios oncology portfolio in 2021.

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36. Anvumetostat (AMG 193) — MTA-cooperative PRMT5 inhibitor

Aliases: anvumetostat, AMG 193, AMG-193

Modality: small_molecule · **Regulatory:** Development discontinued. Amgen announced on its Q1 2026 earnings call (April 30, 2026) that it stopped four phase 1b/2 trials of anvumetostat because efficacy did not meet the company's bar for additional internal investment. · **Guidelines:** Not in any guideline. · **Geographic scope:** Globally discontinued. · **Verified:** 2026-05-26

Amgen discontinued anvumetostat in April 2026 after the company concluded that the cross-tumor efficacy didn't justify continued investment. The Part 1m GBM cohort generated the published phase 1 evidence (Rodon 2024 / 2025) that informed the broader MTA-cooperative PRMT5 field, but the drug itself is no longer accessible. For this patient, the practical path forward is one of the other MTA-cooperative PRMT5 trials — BGB-58067 (Phoenix), MRTX1719 (BMS), or navlimetostat (BMS MountainTAP-5 when it opens).

Next steps

1. Do NOT pursue anvumetostat enrollment — the program is closed.
2. Treat anvumetostat as mechanistic evidence only; redirect the PRMT5-axis effort to BGB-58067 (NCT07485049 — best fit), MRTX1719 (NCT05245500 — confirm GBM eligibility), or navlimetostat (NCT07492680 — opens July 2026 with explicit GBM cohort).
3. If the user wants confirmation, call Amgen medinfo at 1-800-772-6436 — they will state the discontinuation date and any residual compassionate-use availability for previously enrolled patients.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT05094336	1/2	active not recruiting	no	—

Manufacturer / sponsor contact

- **Company:** Amgen
- **Med info phone:**1-800-772-6436
- **Med info email:**medinfo@amgen.com
- **Product info:**https://www.amgenpipeline.com/
- **Compassionate / expanded access:** https://www.amgen.com/responsibility/healthy-people/access-to-medicines/access-to-investigational-medicines
- **Notes:** Amgen medical information (1-800-77-AMGEN) can confirm program status and discuss whether any compassionate-use access remains for previously enrolled patients. Given the program-wide discontinuation, expanded-access for new patients is essentially closed.

Payer / coverage: Not applicable — no commercial product.

Notes: Discontinuation disclosed Amgen Q1 2026 earnings call, 30 April 2026. The decision affected four phase 1b/2 trials across NSCLC, MTAP-null solid tumors, thoracic, and GI/biliary/pancreatic cohorts.

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37. Depatuxizumab mafodotin (Depatux-M / ABT-414) — EGFR-directed ADC

Aliases: depatux-m, depatuxizumab mafodotin, ABT-414

Modality: antibody_drug_conjugate · **Regulatory:** Development discontinued. AbbVie halted Phase 3 INTELLANCE-1 in 2019 after the IDMC concluded the addition of depatux-m to RT + TMZ did not improve OS over placebo in EGFR-amplified newly diagnosed GBM. All ongoing depatux-m studies were halted at the same time. · **Guidelines:** Not in any guideline. · **Geographic scope:** Globally discontinued. · **Verified:** 2026-05-26

Depatux-M is closed. AbbVie stopped the Phase 3 in 2019 after the IDMC found no OS benefit in EGFR-amplified newly diagnosed GBM. The drug is mechanistically interesting (potent preclinical activity, Reilly 2015) but the negative Phase 3 closed the file. Patient cannot access it.

Next steps

1. Treat depatux-m as a closed door.

2. If the EGFR-amplified biology is still of interest, the active strategy is to look at EGFRvIII / EGFR-directed CAR T (Penn, MGH, UCSF — see separate rows) rather than EGFR-directed ADC.

Manufacturer / sponsor contact

- **Company:** AbbVie
- **Product info:**https://www.abbvie.com/
- **Notes:** AbbVie discontinued the program in 2019. No compassionate use is available.

Payer / coverage: Not applicable.

Notes: The INTELLANCE-1 Phase 3 negative readout is the load-bearing event. The drug had robust preclinical activity in EGFR-amplified models which made the negative clinical translation surprising.

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38. Rindopepimut (CDX-110) — EGFRvIII peptide vaccine

Aliases: rindopepimut, CDX-110, Rintega, PEPvIII-KLH

Modality: peptide_vaccine · **Regulatory:** Development discontinued. Phase 3 ACT IV (newly diagnosed EGFRvIII+ GBM) terminated for futility in 2016; median OS 20.4 mo (rindopepimut) vs 21.1 mo (control). All ongoing studies were closed. · **Guidelines:** Not in any guideline. · **Geographic scope:** Globally discontinued. · **Verified:** 2026-05-26

Rindopepimut is closed. ACT IV failed to extend OS in newly diagnosed EGFRvIII+ GBM, and Celldex shut down the program. The patient cannot access rindopepimut, and the EGFRvIII vaccine strategy as a whole has been replaced by EGFRvIII-directed CAR T cells (Penn, MGH, UCSF — see separate rows).

Next steps

1. Treat rindopepimut as a closed door.
2. Redirect EGFRvIII-axis interest to the CAR T programs.

Manufacturer / sponsor contact

- **Company:** Celldex Therapeutics
- **Product info:**https://www.celldex.com/
- **Notes:** Celldex discontinued the program after ACT IV. At the time of termination, Celldex offered compassionate-use access to patients on the ACT IV experimental arm and to existing compassionate-use recipients — but no new compassionate-use enrollment is available.

Payer / coverage: Not applicable.

Notes: ACT IV's failure was a load-bearing data point for the field — the antigen-escape problem (Sampson 2010) had been a long-standing concern, and the Phase 3 confirmed that single-antigen targeting in GBM has the predicted resistance mechanism. The class moved from monovalent vaccines to multivalent cell therapy.

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39. TNG908 — brain-penetrant MTA-cooperative PRMT5 inhibitor

Aliases: TNG908

Modality: small_molecule · **Regulatory:** Investigational, no approval. Trial terminated. · **Guidelines:** Not in any guideline. · **Geographic scope:** Globally discontinued. · **Verified:** 2026-05-26

TNG908 was Tango's brain-penetrant first-generation PRMT5 inhibitor. Tango terminated NCT05275478 after the GBM cohort failed to show meaningful activity and refocused on the next-generation brain-penetrant TNG456. TNG908 itself is not accessible.

Next steps

1. Treat TNG908 as a closed door.
2. If the user wants to follow the Tango brain-penetrant program, ask clinicaltrials@tangotx.com about TNG456 GBM trial timing.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT05275478	1/2	terminated	no	—

Manufacturer / sponsor contact

- **Company:** Tango Therapeutics
- **Med info phone:**857-320-4899
- **Med info email:**clinicaltrials@tangotx.com
- **Product info:**https://www.tangotx.com/
- **Notes:** Same Tango contact as TNG462. The TNG908 program is over; the active Tango program for brain-penetrant PRMT5 inhibition is TNG456.

Payer / coverage: Not applicable.

Notes: The TNG908 termination is informative — it suggests the first-generation brain-penetrant PRMT5 inhibitor class may need optimization (the BGB-58067 and TNG456 successors are the bet) before delivering CNS efficacy. Useful context when evaluating the BGB-58067 Phoenix trial.

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40. Toca 511 + Toca FC (vocimagene amiretrorepvec + 5-fluorocytosine)

Aliases: Toca 511, Toca FC, vocimagene amiretrorepvec, 5-fluorocytosine

Modality: gene_therapy · **Regulatory:** Development discontinued. Phase 3 Toca 5 trial (NCT02414165) failed primary OS endpoint in 2019 (11.1 vs 12.2 mo). Tocagen restructured (~65% staff reduction) and effectively closed the program. The continuation protocol NCT04327011 supported only previously-enrolled patients. · **Guidelines:** Not in any guideline. · **Geographic scope:** Globally discontinued. · **Verified:** 2026-05-26

Toca 511 is closed. The Phase 3 failure in 2019 ended the program. Patient cannot access it; the only remaining role for Toca 511 in the dossier is as mechanistic evidence that prodrug-activator gene therapy didn't translate from the encouraging Phase 1 (Cloughesy 2018 Sci Transl Med) to the registrational Phase 3.

Next steps

1. Treat Toca 511 as a closed door.
2. If the user is interested in gene-therapy approaches to GBM, redirect to the oncolytic-virus programs (CAN-3110, PVSRIPO) which are conceptually distinct but operationally closer to current development.

Manufacturer / sponsor contact

- **Company:** Tocagen Inc. (effectively closed post-2019 restructuring; assets reorganized into Forte Biosciences)
- **Notes:** Tocagen ceased oncology operations after the Toca 5 failure. No commercial or compassionate-use access remains.

Payer / coverage: Not applicable.

Notes: Toca 5 (NCT02414165) was the registrational Phase 3. The Phase 1 (Ostertag 2011 / Cloughesy 2018) had been promising but did not replicate. A frequently-cited cautionary example for retroviral gene therapy in solid tumors.
